

Large pan-cancer cell screen coupled to (phospho-)proteomics underscores high-dose vitamin C as a potent anti-cancer agent

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Abstract

Increasing preclinical and clinical evidence has positioned high-dose vitamin C as a promising anti-cancer treatment that merits more clinical attention. Multiple cytotoxicity mechanisms have been described, including pro-oxidant effects. To contribute to the preclinical understanding of the broad pan-cancer effects of high-dose vitamin C in a global manner, we determined the IC₅₀ of a large panel of cancer cell lines (n=51) representing 7 solid tumour types and generated proteome data. The majority of cell lines were highly sensitive (IC₅₀ range 0.036-10mM, mean 1.7 ± 0.4 mM), well below a clinically achievable dose. The proteome data (>5000 proteins per sample), showed that high sensitivity is associated with proliferation, as indicated by functional enrichment of cell cycle, RNA splicing and chromatin organization, while lower sensitivity is linked to extracellular vesicles, glycolysis, fatty acid metabolism and mitochondria. Moreover, (phospho-)proteome analysis of on-treatment vitamin C effects on four pancreatic ductal adenocarcinoma (PDAC) cells dosed at a range of IC₅₀ values (Hs766 T, 2 mM; Capan-2, 0.6 mM; PANC-1, 0.14 mM and Suit-2, 0.1 mM) revealed, next to cell line specific effects, down-modulation of AKT-MTOR signalling and immune suppressive signalling, while IFN- α response was enhanced upon vitamin C. Altogether, our comprehensive pharmacological and (phospho-)proteome analysis is the first to assess cancer vulnerabilities and effects of vitamin C on a large cancer cell line panel and underscores the potential of high-dose vitamin C as an anti-cancer agent.

eLife assessment

This **useful** study utilizes proteomics analysis across a large panel of 51 cancer cell lines to elucidate mechanisms underlying the sensitivity of cancer cells to high-dose vitamin C (Ascorbate). While the associations between specific molecular pathways and sensitivity to ascorbate are interesting, a major limitation is that the study is largely descriptive and **incomplete**, lacking evidence on the molecular underpinnings of cancer cells' sensitivity to high-dose vitamin C.

1. INTRODUCTION

Vitamin C (VitC) is a natural compound that has prooxidant and anticancer properties when administered intravenously in high doses(1–5). The prooxidant properties are mainly due to the autoxidation of VitC, which forms hydrogen peroxide (H_2O_2) and reactive oxygen species (ROS) (2,6,7). High-dose VitC can specifically target cancer cells due to their unique vulnerabilities compared to normal cells. Cancer cells have higher levels of labile iron, which can increase the formation of hydroxyl radicals(1,8). They often have defective mitochondria and a higher metabolic rate resulting in endogenously elevated levels of oxidative stress(9). VitC can further increase these oxidative stress levels and thereby promote cancer cell death and inhibit metastatic dissemination(10). Moreover, the activity of catalase, an enzyme that breaks down H_2O_2 , is often reduced in cancer cells, allowing H_2O_2 to accumulate and form hydroxyl radicals(1,3,11). Healthy cells, on the other hand, are equipped with molecular features to reduce oxidative stress induced by high-dose VitC.

Upon cell uptake, and in addition to its prooxidant and reducing capabilities, VitC can induce oxidative stress damaged DNA or reverse epithelial-to-mesenchymal transition(12–14). Other research suggests that VitC increases oxidative stress by inducing overexpression of tumour protein P53 in cancer cells(1). Moreover, VitC has been found to inhibit the ROS-dependent MAPK/ERK signalling pathway(15,16), which is crucial for cell growth and is particularly important for proliferation through ERK. This mechanism has been observed in other cancer types, further underscoring the diverse anticancer properties of VitC(17–20).

Accumulating evidence shows the therapeutic potential of high-dose VitC in both preclinical and clinical settings(1). Importantly, high-dose VitC has been widely shown to sensitize cancer cells to many conventional chemo- and radiation therapies and more recently to enhance cancer immunotherapy effects in multiple cancer types *in vivo*(21). In the clinical setting, the use of high-dose VitC treatment, either alone or in conjunction with other therapies such as chemo- and/or radiotherapy, can produce notable anti-cancer effects, as shown in phase I/II studies and case reports. Especially in the palliative setting, the highly safe and tolerable high-dose VitC profile makes it a promising alternative for patients who typically have limited treatment options and a poor prognosis. Finally, it has been shown to significantly reduce the side effects caused by standard therapies in breast cancer and bone metastasis patients(22,23).

Large-scale pharmacological testing in conjunction with global profiling is of high interest to more fully understand the biology associated with sensitivity to high-dose VitC. System-wide approaches such as transcriptomic and proteomic studies can help capture the complexity of various cellular signalling pathways in an unbiased manner. Current transcriptome and proteome studies of high-dose VitC treatment are limited to a few small scale cell line studies(1,3,24). To our knowledge, this study is the first to perform large scale pharmacological analysis coupled to (phospho-)proteomics profiling. We report the sensitivities of VitC in 51 cell lines across 7 major

cancer types that are all in the clinically achievable range, and identify proteins and pathways associated with VitC sensitivity. To further elucidate the mechanism of action of VitC, we profiled the (phospho-)proteomes of four PDAC cell lines treated with VitC at different doses and time points. Our data support further exploration of high dose VitC in the clinical setting.

2. MATERIALS AND METHODS

2.1. Cell lines and culture conditions

A total of 51 immortalized human cancer cell lines representative of pancreatic, lung, prostate, breast and colorectal cancer were used in this study. Cells were cultured with Dulbecco's modified Eagle's medium (DMEM) or Roswell Park Memorial Institute (RPMI) 1640 Medium following manufacture recommendations, and supplemented with 10% fetal calf serum (FCS) (Biowest) and 1% penicillin/streptomycin (P/S). All cell lines were cultured at 37°C in 5% CO₂.

2.2. Preparation of 1M VitC stock solution

The 1M stock solution of VitC was freshly made for each experiment. VitC (05878 Fluka, Honeywell) was dissolved in type 1 MilliQ water (18 MW). The pH was then adjusted with 10M and 1M sodium hydroxide solution to pH 7.0-7.4. After adjusting the pH, the solution was filtered using a sterile syringe filter coupled to a filter unit with a low protein binding Durapore PVDF membrane (Merck Millipore) and the solution was poured into an amber tube to protect from light exposure.

2.3. Determination of cell line vitamin C sensitivity

Cells were trypsinized with 10x diluted Trypsin-EDTA (0.5%) (Gibco) solution for 5 minutes until detachment. Once trypsin was removed by centrifugation, cells were subsequently counted by using the LUNA-II™ Automated Cell Counter. Cells were then seeded in 96 well flat bottom Eppendorf plates at pre-determined densities. After 24 hours, allowing a complete adherence of the seeded cells, fresh 1M stock of VitC was prepared.

Forty out of fifty four cell lines were then dosed per cell by counting them prior to treatment. This was done to adjust concentrations to pmol/cell based on their doubling time to ensure reproducibility as shown in previous studies(25,26). Average number of cells in the well prior to treatment (X) and expected number of cells after 24h (Y) were then used to adjust pmol/cell concentrations using the following formula: VitC concentration pmol/cell = X cells in 100 µL / Y * treatment volume. Cells were then treated for 2h with pharmacologic concentrations of VitC in a range of 0-20 mM. Respective dilutions were made with the corresponding media. After treatment, medium was carefully removed and replaced with fresh medium. A total of eleven cell lines were not dosed per cell but were included in the analysis using standard mM dosing.

Cell growth was determined by Sulforhodamine B (SRB) assay(25,26) after 72 hours of treatment at 37°C and 5% CO₂. In brief, a total of 25 µl of cold 50% Tri-chloroacetic acid (TCA) were added to each well and incubated for at least 60 min at 4°C. Plates were then washed five times with demi water and dried at room temperature (RT). 50 µl of SRB solution (0.4 % Sulforhodamine B (Sigma-Aldrich) and 1% acetic acid (Thermo Fisher Scientific) were subsequently added per well and cells were stained for 15 min at RT. Plates were then washed four times with 1% acetic acid solution and dried at RT. Finally, 150 µl of 10 mM Tris solution were added per well and plate was vortexed for 5 min. Data were read out by a Microplate Reader (Biotek, Synergy HT) at 490/540 nm. VitC sensitivity of all cell lines was determined by calculating the IC₅₀s using pmol/cell and mM units. Since the SCLC cell line panel included suspension and semi-adherent cell lines, cell viability was determined using CellTiter-Fluor instead of SRB assay. Dose-response curves were generated and IC₅₀ values calculated using GraphPad Prism software.

For catalase assay, cells were co-exposed with catalase (50U/ml) and VitC for 2h. Catalase was added 1 min prior to exposure as previously shown(28). Confluence images were taken by the Essen Bioscience IncuCyte™ ZOOM system and software.

2.4. Determination of cell line cisplatin sensitivity

Viability assays were performed to determine the IC50 values of cisplatin for the LUAD cell line panel. Depending on each cell lines' proliferation time, 3000-5000 cells were seeded in 96 well flat bottom Eppendorf plates, left to attach for 24h and subsequently treated with a range of different cisplatin concentrations (0-1000 μ M; Accord Healthcare) for 72h. Cell growth was determined using the SRB assay. Dose-response curves were generated and IC50 values calculated using GraphPad Prism software.

2.5. Intracellular ROS levels

Intracellular ROS levels were measured in Capan-2 and PANC-1 cell lines after 10 minutes, 2h and 24h of VitC treatment by using the OxiSelect™ Intracellular ROS Assay Kit (Green Fluorescence). Fluorescence was measured with a Biotek plate reader at 480 nm/530 nm wavelengths.

2.6. Preparation of untreated (baseline) cell lysates

Cells were grown up to ~70% confluency in 10 cm petri dishes. Plates were washed with PBS and subsequently lysed in 1x SDS PAGE sample buffer. If the lysate was too viscous after scraping, 50-100 μ L additional 1x SDS PAGE sample buffer was added. Next, cell lysates were sonicated using a Branson Digital Sonifier for 1 min. in 3 cycles of 20" on/20" off and then centrifugated for 10 min at max. speed. The supernatant was transferred to a new Eppendorf tube and stored at -20°C.

2.7. Preparation of on-treatment PDAC cell line lysates

For on-treatment (phospho-) proteome profiling, four PDAC cell lines (Suit-2, PANC-1, Capan-2 and Hs 766t) were seeded in 15 cm dishes and cultured until 60% confluency. Then, cell lines were treated with respective VitC IC50 doses (0.1, 0.14, 0.6 and 2 mM, respectively), and lysates were collected in a time series of 2, 4 and 24 hours post-treatment. For each time point and cell line, a control with no treatment was included (Fig. 9E). Furthermore, each condition consisted of two technical replicates, except for 2h- and 4h-control dishes, where each time point had one replicate due to the short time difference. After 2h of treatment, the media of the 4h and 24-hour dishes was refreshed with media without VitC, as previously described(7,27). This was done to maintain the influx of H2O2 and to reduce the influence of breakdown products of VitC.

For each time point and cell line, we isolated the phospho-protein fraction as previously described(28). In brief, the supernatant was discarded and cells were washed with cold PBS at least three times. Immediately after, urea lysis buffer (9 M urea, 20 mM HEPES pH 8.0, 1 mM sodium orthovanadate, 2.5 mM sodium pyrophosphate, 1 mM β -glycerophosphate) was added and cells were removed from the substrate with a cell scraper. Lysate was collected in a 15 ml tube and stored at -80°C until sonication. The lysates were then subjected to sonication (3 cycles of 20" on/20" off at 18 micron amplitude) using a MSE Soniprep 150 sonicator and centrifugation (24,000G, 15 minutes, 10 °C). Supernatant was transferred to a new tube, aliquoted and stored at 80°C until phospho-peptide capture. To quantify protein concentration, the Pierce BCA Protein Assay kit (Thermo Scientific) was used.

2.8. 5-band in-gel digestion

To determine optimal and comparable loading amounts for each untreated (baseline) cell line, different amounts of protein lysate were first loaded onto 12.5% SDS-PAGE test gels and proteins separated using the BioRad SDS-PAGE system (Biorad, Hercules, CA). Gels were run at 150 V for 50 min., fixed in 50% ethanol/3% phosphoric acid solution and stained with Coomassie brilliant blue

G-250. Relative lysate amounts were quantified using ImageJ software. Subsequently, protein lysates were loaded onto a precast 4-12% gradient SDS-PAGE gel from the NuPAGE SDS-PAGE system (Invitrogen, Carlsbad, CA). The gel was run at 200 V for 40 min., fixed in 50% ethanol/3% phosphoric acid solution and stained with Coomassie brilliant blue G-250. Subsequent in-gel digestion was performed as described before(29). In brief, proteins were reduced using 10 mM DTT and alkylated using 50 mM iodoacetamide. Then, in a laminar flow cabinet, each gel lane was cut into 5 bands, and each of them cut into ~ 1 mm³ cubes, which were transferred to a pre-labelled Eppendorf tube. Every band was then separately digested overnight using trypsin. The peptides extracted from each gel band were subsequently stored at -20°C until LC-MS/MS analysis.

2.9. Sample preparation for phospho-proteomics

All lysates were reduced, alkylated and digested with Sequencing Grade Modified Trypsin (Promega, USA). For digestion and desalting an equivalent of 500 µg protein was used. Upon lysis proteins were reduced with DTT (dithiothreitol, 5 mM) and alkylated with IAA (iodoacetamide, 10 mM) followed by dilution to 2 M urea. Samples were digested with trypsin overnight (1:100 m/m) and then acidified by addition of 0.1% Tri-fluoroacetic acid (TFA). The digests were desalted using Oasis HLB columns (500 mg capacity, Waters), eluted in 0.1% TFA, 80% acetonitrile (ACN) and lyophilized for 48 h in a freeze dryer (Christ alfa 2-4 LSC).

Phospho-peptide enrichment was performed by IMAC (immobilized metal ion affinity chromatography), using the BRAVO AssayMap liquid handling robot (Agilent) as previously described(28). In brief, 200 µg peptides for each sample were acidified to 0.1% TFA, and applied to a 10 mg OASIS HLB cartridge (Waters) previously activated with acetonitrile and equilibrated with 0.1% TFA. After washing the columns twice with 0.1% TFA, the bound phospho-peptides were eluted in 100 µl 5% NH₄OH/30% ACN and then transferred to glass-lined autosampler vials and dried in a vacuum centrifuge at 45 °C. Finally, samples were re-dissolved in 20 µl 0.5% TFA/4% ACN prior to injection and stored at 4 °C until LC-MS/MS analysis. Two replicates of the control lysate of HCT116 (colon carcinoma cell line) were included in the processing as workflow control.

2.10. LC/MS-MS Analysis

For the untreated (baseline) proteomic profiling, peptides were separated by an Ultimate 3000 nanoLC-MS/MS system (Dionex LC-Packings, the Netherlands) equipped with a 40 cm × 75 µm ID fused silica column custom packed with 1.9 µm 120 Å ReproSil Pur C18 aqua (Dr Maisch GMBH, Germany). After sample injection, peptides were trapped at 6 µl/min on a 10 mm × 100 µm ID trap column packed with 5 µm 120 Å ReproSil Pur C18 aqua at 2% buffer B (buffer A: 0.5% acetic acid (Fischer Scientific), buffer B: 80% ACN, 0.5% acetic acid) and separated at 300 nl/min in a 10–40% buffer B gradient in 90 min (120 min inject-to-inject) at 35 °C. The eluting peptides were ionized at a potential of + 2 kVa into a Q Exactive mass spectrometer (Thermo Fisher, Germany). Intact masses were measured at resolution 70,000 (at m/z 200) in the Orbitrap using an AGC target value of 3 × 10⁶ charges. The top 10 peptide signals (charge-states 2+ and higher) were submitted to MS/MS in the HCD (higher-energy collision) cell (1.6 amu isolation width, 25% normalized collision energy). MS/MS spectra were acquired at resolution 17,500 (at m/z 200) in the orbitrap using an AGC target value of 2 × 10⁵ charges and an underfill ratio of 0.1%. Dynamic exclusion was applied with a repeat count of 1 and an exclusion time of 30 s.

For the on-treatment phospho-proteomic profiling, peptides were separated on an Ultimate 3000 nanoLC-MS/MS system (Dionex LC-Packings, Amsterdam, the Netherlands) equipped with a 50-cm 75 µm ID C18 Acclaim pepmap column (Thermo Scientific). After injection, peptides were trapped at 3 µl/min on a 10-mm, 75-µm ID Acclaim Pepmap trap column (Thermo Scientific) in buffer A (buffer A: 0.1% formic acid, buffer B: 80% ACN/0.1% formic acid), and separated at 300 ml/min with a 10–40% buffer B gradient in 90 min (120 min inject-to-inject). Eluting peptides were ionized at a potential of +2 kV and introduced into a Q Exactive HF mass spectrometer (Thermo Fisher, Bremen, Germany). Intact masses were measured in the orbitrap with a resolution of 120,000 (at

m/z 200) using an automatic gain control (AGC) target value of 3×10^6 charges. Peptides with the top 15 highest signals (charge states 2+ and higher) were submitted to MS/MS in the higher-energy collision cell (1.6-Da isolation width, 25% normalized collision energy). MS/MS spectra were acquired in the Orbitrap with a resolution of 15k (at m/z 200) using an AGC target value of 1×10^6 charges and an under fill ratio of 0.1%. Dynamic exclusion was applied with a repeat count of 1 and an exclusion time of 30 s.

2.11. Phospho-protein identification

MS/MS spectra were searched against the Swissprot human reference proteome 2020_04_canonical_and_isoform.FASTA (42347 entries) using MaxQuant(30) 1.6.10.43. Enzyme specificity was set to trypsin and up to two missed cleavages were allowed. Cysteine carboxamidomethylation (+ 57.021464 Da) was treated as fixed modification and methionine oxidation (+ 15.994915 Da) and N-terminal acetylation (+ 42.010565 Da) as variable modifications. Peptide precursor ions were searched with a maximum mass deviation of 4.5 ppm and fragment ions with a maximum mass deviation of 20 ppm. Peptide and protein identifications were filtered at an FDR of 1% using the decoy database strategy. The minimal peptide length was 7 amino-acids and the minimum Andromeda score for modified peptides was 40 and the corresponding minimum delta score was 6. Proteins that could not be differentiated based on MS/MS spectra alone were grouped to protein groups (default MaxQuant settings). Protein expression searches were performed with the label-free quantification option selected, and match between runs option was off. Proteins were quantified using spectral count MS data, which was subsequently normalized by a global normalization to the total count of each sample.

MS/MS spectra of IMAC phosphopeptide enrichment experiments were searched separately against the Swissprot human FASTA file (canonical and isoforms; 2021, 42383 entries) using MaxQuant 1.6.10.43. Enzyme specificity was set to trypsin, and up to two missed cleavages were allowed. Cysteine carboxamidomethylation (+57.021464 Da) was treated as fixed modification and serine, threonine, and tyrosine phosphorylation (+79.966330 Da), methionine oxidation (+15.994915 Da), and N-terminal acetylation (+42.010565 Da) as variable modifications. Peptide precursor ions were searched with a maximum mass deviation of 4.5 ppm and fragment ions with a maximum mass deviation of 20 ppm. Peptide and protein identifications were filtered at a false discovery rate of 1% using a decoy database strategy. The minimal peptide length was set at 7 amino acids, the minimum Andromeda score for modified peptides was 40, and the corresponding minimum delta score was 6. Proteins that could not be differentiated based on MS/MS spectra alone were clustered into protein groups (all default MaxQuant settings). Phosphopeptide identifications were propagated across samples using the 'match between runs' option checked. For phosphopeptide data, we used data from the MaxQuant 'modificationSpecificPeptides' table. For phosphosite data, we used data from the MaxQuant 'Phospho (STY) Sites' table. Phosphopeptides and phosphosites were quantified from the area under the curve of the MS1 signal of each eluting peptide ('Intensity' in MaxQuant). Human Swiss-Prot database used for raw data search was acquired from the UniProt database (<https://www.uniprot.org/>). For calculation of kinase INKA scores, phosphopeptide MS/MS spectral counts were calculated from the MaxQuant evidence file using R.

2.12. Statistical analysis and pathway annotation

To unravel biology and proteins associated to VitC sensitivity, and considering the unique molecular characteristics of each cancer type, we performed two complementary statistical analyses per panel: a Spearman correlation and a two-group comparison. Spearman correlation analyses were performed between VitC half-maximal inhibitory concentrations (IC50s) and the protein expression for the whole panel, using the R package *stats*. Group comparisons with proteomic data (2 lowest IC50 vs 2 highest IC50, or 4 vs 4 cell lines, depending on panel size) were performed by *bb* test using the *ion* package in R. For proteins correlating with VitC IC50 values, a filtering threshold of $\text{Rho} \geq 0.7$ and a *pvalue* of ≤ 0.05 was the standard for the majority of panels,

except for CRC for which the Rho threshold was set to 0.6 to increase number of identifications. Two group comparison analyses were performed on the cell lines with lowest and highest VitC sensitivity per panel, whereby number of samples in low and high sensitivity groups differed per panel depending on panel size. Filtering threshold was set to a fold change of ≥ 2 and a *pvalue* of ≤ 0.05 . The overlap between the two statistical analyses was performed on the significant candidates associated with low and high sensitivity, and the enriched biology was further investigated using gProfiler(31). Top 10 enriched and non-redundant biology terms were selected. A more stringent analysis was followed to select top 20 proteins associated with high and low sensitivity, including only significant hits which had a Rho threshold of ≥ 0.7 and a fold change of ≥ 1.5 . Top candidates involved in known VitC associated processes were further annotated, as well as those that were only significant in one analysis but involved in these processes.

For gene set enrichment analysis (GSEA), significant proteins were ranked by multiplying the sign of the fold change by the $-\log_{10}$ *pvalue*. This was used as input for GSEA(32) using the *fgsea*(33) package in R. Gene signatures were retrieved with the *msigdb*(34) package. Single sample gene set enrichment analysis (ssGSEA) was performed using the GSEA R package(35). For the targeted analysis, known processes associated with VitC sensitivity(1,3) were retrieved from Reactome(36) and Gene Ontology(37,38) of biological and molecular function processes.

For the phospho-proteome data analyses, we used the log-transformed median-site normalized intensities. To determine differences between VitC and control samples, fold changes (FC) were subsequently calculated with the normalized intensities. In brief, averaged VitC treated replicates were divided by the averaged controls per timepoint to calculate upregulation, and -1 times the averaged controls divided by averaged VitC treated replicates per timepoint was used to calculate downregulation. These calculations resulted in FC where 1 and -1 were equivalent to no change between VitC treated samples and controls. Kinase activity was inferred from phospho-proteomic data with the Integrative iNferred Kinase Activity (INKA) tool as previously described by Beekhof et al. (39) and available at <https://inkascore.org/>. The kinases were ranked according to the sum of all INKA scores per kinase of all samples divided by the number of samples. Based on the ranking, the top 20 ranked kinases and boxplots of the INKA scores were made. Fold changes were then calculated per cell line and timepoint with the same method as described above and heatmaps with the corresponding FC were made per cell line.

3. RESULTS

3.1. High-dose VitC pan-cancer sensitivities

To further underscore the potential of high-dose VitC as a broadly acting anti-cancer agent, we determined VitC responses in a large cancer cell line panel containing seven major cancer types (PDAC, LUAD, LUSC, SCLC, CRC, BC and PC, Fig.1A).

Pharmacological VitC testing was performed in all cell lines, which were treated with ten VitC concentrations (0-20 mM), and the respective sensitivity was inferred by using IC50s at mM and pmol/cell dosing (Fig. 1C-D, respectively). For the majority of cell lines, a 5 mM VitC dose was sufficient to inhibit 50% of cell growth, and among all, average IC50 was 1.7 ± 0.4 mM (range 0.036 - 10.02 mM). Due to the known VitC and H₂O₂ toxicity dependency on cell density(2,40), a dosing per cell metric scheme was applied for a selection of cell lines (n=40 in PDAC, LUAD, LUSC, PC and BC panels) and an average of 18.7 ± 3 pmol/cell (range 0.95-63.2 pmol/cell) was observed (Fig. 1D). Interestingly, the two metrics -mM and pmol/cell-showed comparable results to a large extent (Fig. 1C-D, Pearson coefficient $R=0.892$, *pvalue*= 1.143×10^{-13}).

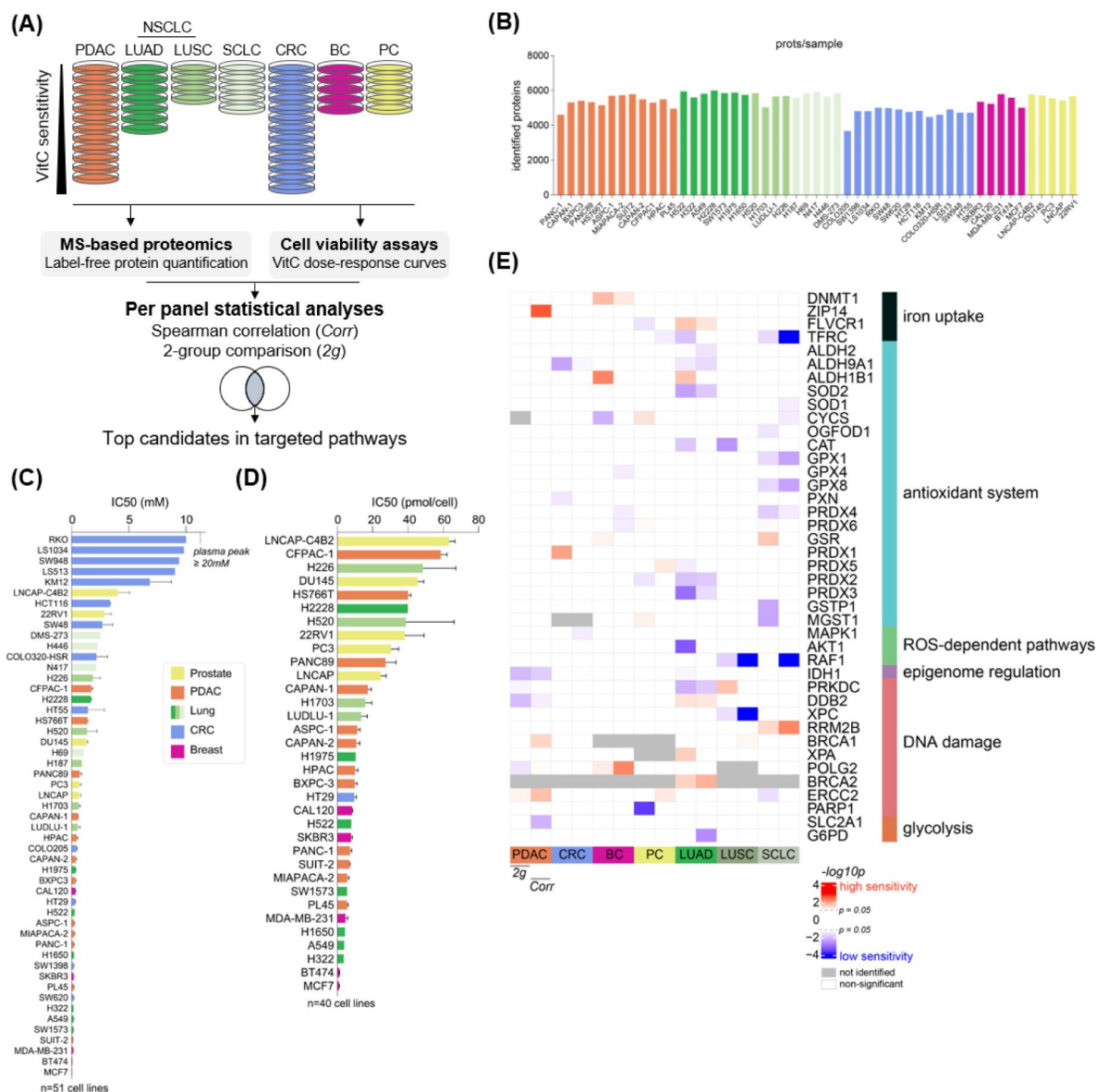


Fig. 1.

Pan-cancer proteomic analysis of vitamin C sensitivity.

(A) Study design of the pan-cancer proteome profiling of 51 human cell lines divided by cancer (sub)type panel: PDAC, LUAD, LUSC, SCLC, CRC, BC and PC, with the determination of vitamin C sensitivities. The candidate selection of sensitivity-associated proteins per panel was supported by two complementary statistical analyses: a Spearman correlation of protein expression with IC50 values (Corr), and a two-group comparison between protein expression in more and less sensitive cell lines (2g). (B) Number of identified proteins per sample. (C) (D) VitC sensitivities plotted as IC50s, measured by cell viability assays and represented (C) in millimolar (n=51 cell lines) and (D) in pmol/cell range (n=40 cell lines). Mean $\pm$ SEM plotted, average $n \geq 2$. (E) Targeted analysis of potential VitC-associated processes⁽¹⁾. The significance by two group comparison and Spearman correlation is shown per panel. Proteins with at least one significant *p*value in at least one panel were included.

Altogether, our large scale pan-cancer pharmacological analysis showed that all cancer cell lines were killed at clinically achievable concentrations (below two-fold the 20 mM plasma peak), extending the finding of previous smaller scale cancer cell line studies(1 [43](#),41 [44](#)) and underscoring its promise as a broad anti-cancer therapy.

3.2. Per cancer cell line panel analysis of biological processes and proteins associated with high-dose VitC sensitivity

Little is known about the proteome association with high-dose VitC cancer vulnerabilities. In fact, to our knowledge, the present study is the first to evaluate the baseline proteomic features associated with high-dose VitC responses in a large pan-cancer panel. To this end, polyacrylamide gel fractionation coupled to MS-based label-free proteomics was applied to the 51 cancer cell lines for which we determined the VitC IC50 values. On average, 5352 proteins were identified per sample (**Fig. 1B** [45](#)). Because these cell line panel datasets were generated at different points in time, we analysed the data per panel.

First, we evaluated the expression of several proteins of processes previously found to be associated with VitC sensitivity such as iron uptake, antioxidant defence system, DNA repair or glucose metabolism(1 [43](#),3 [46](#)) (**Fig. 1E** [47](#)). Two complementary statistical approaches were applied to each panel and proteins significant in either statistical analyses, 2-group comparison or Spearman correlation, were evaluated. As expected, proteins involved in the antioxidant system, such as the renowned ROS scavengers CAT, SOD1/2, GPXs and PRDXs, were significantly increased in less sensitive cell lines from multiple cancer types, particularly LUAD and SCLC. Similarly, ROS-related pathways(42 [48](#)) were found high in less sensitive cell lines. Increased intracellular iron, in the Fe²⁺ form, reacts with hydrogen peroxide to form hydroxyl radicals, thus damaging DNA, proteins and lipids, and leading to cancer cell death, thereby increasing VitC sensitivity. Iron transporter TFRC expression, that mediates transport of iron bound to transferrin, was found high in less sensitive PC, LUAD and SCLC cell lines. However, proteins involved in “free” iron transport and mobilization(43 [49](#),44 [50](#)), e.g. DNMT1, ZIP14 and FLVCR1, were found to be significantly increased in sensitive cell lines, especially in the BC, PDAC and LUAD panels. Increased abundance of glucose transporter SLC2A1 (GLUT1), which is known to transport VitC into the cell, was found to be significant in less sensitive PDAC cell lines. Glycolysis-related protein G6PD, which is also an oxidative stress protector, was also found significantly high in less sensitive LUAD cell lines. On the contrary, DNA damage/repair proteins such as DDB2, RRM2B, BRCA1, XPA and POLG2 were found to be particularly elevated in cells with high sensitivity, which may be also linked to proliferation signals. IDH1, regulator of DNA methylation, was high in less sensitive cell lines in the PDAC and LUAD panels.

For functional data mining and selection of VitC sensitivity-associated proteins, the intersection of the 2-group and correlation statistical analyses was used. Below we present the results per cancer panel.

For one of the deadliest cancer types, PDAC, we investigated a large panel containing 12 PDAC cell lines, which displayed a diverse sensitivity range (5.5-58.6 pmol/cell and 0.12-1.72 mM, **Fig. 2A-C** [51](#)). Of note, no association between sensitivity and the epithelial/mesenchymal phenotypes was found (**Fig. 2A** [52](#)), which underscores the one-size-fits-all capability of VitC as an anti-cancer agent. Proliferative functions such as cell cycle, RNA metabolism, translation and mitochondrial processes were enriched in highly sensitive cell lines (**Fig. 2D** [53](#)). High-sensitivity top 20 candidates were found to be mainly involved in RNA splicing (e.g. SNRNP200, SRRM2 and AQR) and RNA translation and transcription (e.g. GTF3C1, RPS5, RPS7, RRP12, SIN3A, BMS1, GTF3C5 and SMG1). In addition, nuclear pore complex protein TPR, which is also involved in glucokinase transport, and protein repair protein PCMT1 were among the top 20 high sensitivity candidates (**Fig. 2F** [54](#)). Cells with lower sensitivity displayed higher cell adhesion, extracellular vesicle, and mevalonate pathway functions, the latter being involved in cholesterol synthesis and protein

prenylation (**Fig. 2E**). Among the top 20 candidates associated with lower VitC sensitivity were the superoxide producer CYBA, DNA damage repair proteins UBE2V1, PPP5C and POLB, calcium-ion binding proteins S100A6, S100A16, S100A13 and CIB1, the cholesterol synthesis protein MVD, and two important glycolysis and TCA proteins, PFKL and MUT, respectively (**Fig. 2G**).

Likewise, we investigated a large panel containing 13 colorectal cancer (CRC) cell lines, which displayed a sigmoid sensitivity range (0.16-10 mM, **Fig. 3A-C**). CMS subtypes did not correlate with VitC sensitivity, confirming the universal anti-cancer effect of VitC, regardless of molecular subtype, as already observed in the PDAC panel. Processes associated with high VitC sensitivity were again linked to proliferation including mRNA metabolism and splicing, cell cycle and DNA damage responses (**Fig. 3D**). Interestingly, a great number of pre-mRNA splicing proteins were found among the top 20 candidates, such as SF3A1, LUC7L2, PUF60, TRA2B, PRPF38A and SRSF10, as well as transcription regulators HNRNPH1, CNBP, ATRX, EIF4H, RAE1 and SMARCD1 (**Fig. 3F**). Proteins involved in energy and glucose metabolism such as CKB, VDAC2, AKR1B1 and RAE1 were also indicators of high VitC sensitivity. Processes associated with lower sensitivity were vesicle-mediated transport, glucose metabolism, cell adhesion and protein transport (**Fig. 3E**). Interestingly, among the top 20 proteins were mitochondrial activity related proteins such as HSPA9, HADHA and STOML2, together with other metabolic proteins such as the key player of glycolysis, GPI, lipid synthesis protein ACSL1, and SPR, promoter of BH4, which is a potent radical-trapping antioxidant(45) (**Fig. 3G**). Another significant protein that could influence VitC sensitivity was pyruvate dehydrogenase complex member PDHB, involved in the glycolysis to TCA cycle step.

As for CRC, a sigmoid response was also observed for the breast cancer panel containing 5 cell lines, which showed the highest sensitivity among all panels (0.95-8.4 pmol/cell and 0.036-0.32 mM, **Fig. 4A-C**). All cell lines were thus considered to be relatively sensitive to VitC, regardless of breast cancer subtypes. Biological processes such as DNA replication and DNA damage responses, together with nuclear processes and telomere C-strand synthesis were enriched in the more sensitive cells (**Fig. 4D**). Top high-sensitivity candidate proteins were strongly involved in DNA repair, such as HIST1H2BM, DDB1, TOP2B, HIST2H3A, MDC1, SUPT16H and PDS5A, as well as in RNA splicing, as previously also observed in the PDAC and CRC panels, represented by proteins such as SFPQ, SF3B1, DHX15, DDX46 and THRAP3 in the breast cancer panel (**Fig. 4F**). Significant candidates involved in oxidative stress were also identified: OXSR1, which regulates kinase signalling in response to stress, and QSOX2, a hydrogen peroxide generator, which supports the fact that sensitive cells indeed have boosted levels of ROS(2). Terms enriched in breast cancer cell lines with reduced VitC sensitivity were ribosomal and amino acid metabolism, mitochondrial and extracellular exosome (**Fig. 4E**). Ribosomal-related proteins, e.g. RPL5, RPL7, RPS6, RPL13A, MRTO4 and RPL13, were particularly represented in the top 20 candidates for low VitC sensitivity (**Fig. 4G**). In addition, we identified two-crucial proteins for the mitochondrial beta-oxidation pathway, ACAT1 and ECHS1. Interestingly, we also identified PYCR1, which promotes proline synthesis by promoting glycolysis and nucleotide synthesis, as well as inhibiting ROS levels(48). Nucleotide metabolism-related proteins were also identified (ADK, AHCY and TK1). Two proteins were involved in oxidative stress homeostasis: PHB2, which has recently been shown to be involved in the regulation of mitochondrial ROS levels in glioblastoma(49), and GSTK1, which is an antioxidant contained in peroxisomes. Other significant and interesting proteins were also involved in oxidative stress homeostasis, e.g. LANCL1, which functions as a glutathione transferase, as well as ATP6V0D1, involved in iron homeostasis by triggering Fe²⁺ prolyl hydroxylases, which in turn hydroxylate HIF1A and target it to proteasomal degradation(50). We also identified four mitochondrial proteins, NDUFA9, NDUFS6, NDUFA5, NDUFAF5, which are part of mitochondrial respiratory chain Complex I, associated to low VitC sensitivity.

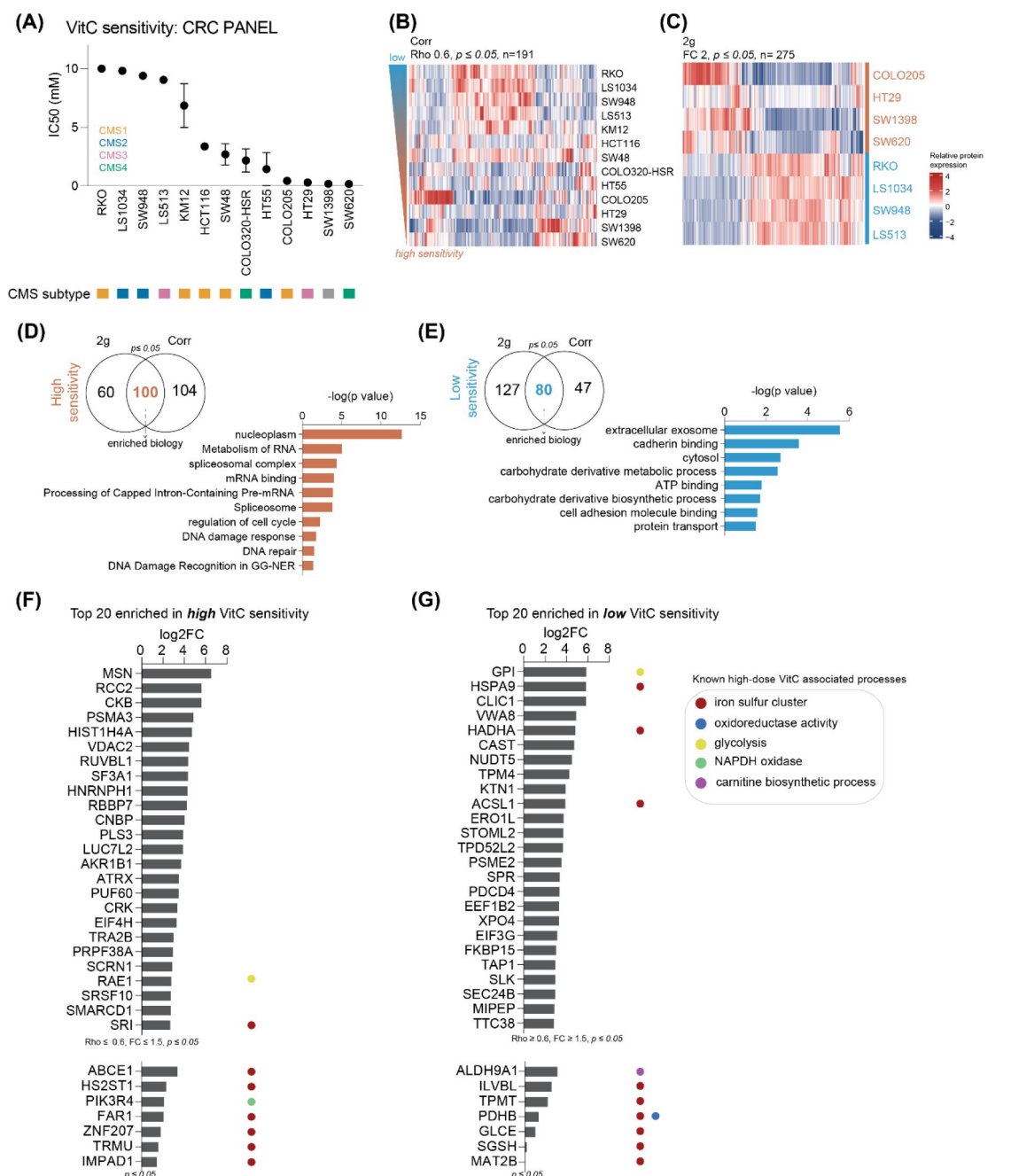


Fig. 3.

Colorectal cancer VitC sensitivity and protein biomarker discovery analysis.

(A) Vitamin C sensitivities (IC₅₀ mM) of $n=13$ CRC cell lines. CMS subtype information was retrieved from two recent studies (46, 47). (B) Hierarchical clustering heatmap of proteins correlated with high and low VitC sensitivity (Spearman Correlation $Rho \geq 0.6$, $pvalue \leq 0.05$, $n=191$ identified proteins). (C) Hierarchical clustering heatmap of proteins enriched in highly sensitive and lowly sensitive cell lines (2-group comparison, $FC \geq 2$, $pvalue \leq 0.05$, $n=275$ identified proteins). (D), (E) Venn diagram describing the overlap between significant proteins found in both statistical analysis, 2-group comparison and Spearman Correlation for (D) high and (E) low VitC sensitivity, as input for the top 10 enriched biology terms. Terms for (D) high and (E) low VitC sensitivity are ranked on $-\log_{10} pvalue$. (F), (G) Top 20 candidate markers associated with (F) high and (G) low VitC sensitivity ranked on log₂FC. Gene ontology terms of known VitC related processes were further annotated. Significant proteins found in these VitC-related processes were also added.

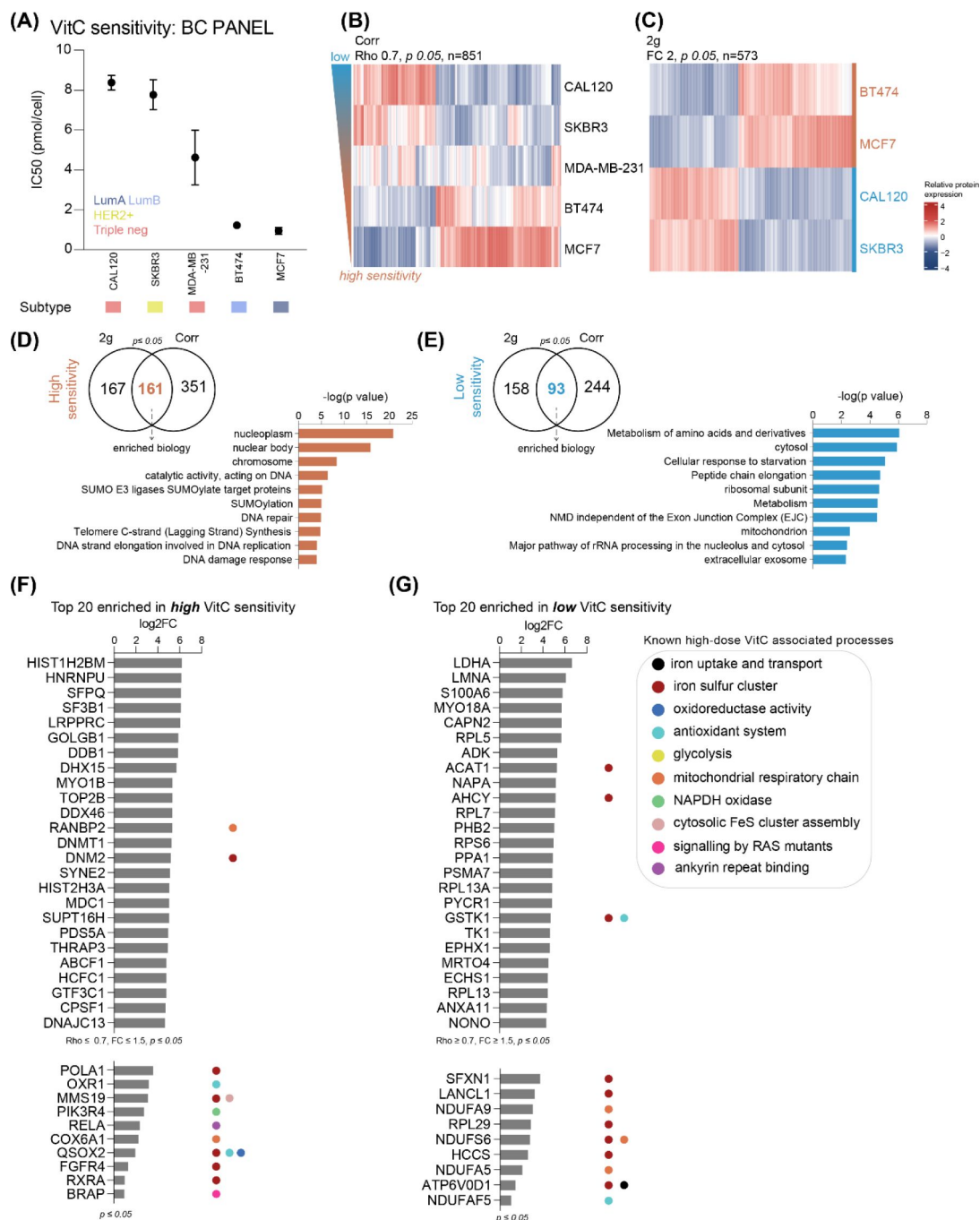


Fig. 4.

Breast cancer VitC sensitivity and protein biomarker discovery analysis.

(A) Vitamin C sensitivities (IC50 pmol/cell) of $n=5$ BC cell lines. Molecular subtype was annotated based on a reference study(51). (B) Hierarchical clustering heatmap of proteins correlated with high and low VitC sensitivity (Spearman Correlation $Rho \geq 0.7$, $p \leq 0.05$, $n=851$ identified proteins). (C) Hierarchical clustering heatmap of proteins enriched in high and low sensitive cell lines (2-group comparison, $FC \geq 2$, $p \leq 0.05$, $n=573$ identified proteins). (D), (E) Venn diagram describing the overlap between significant proteins found in both statistical analysis, 2-group comparison and Spearman Correlation for (D) high and (E) low VitC sensitivity, as input for the top 10 enriched biology terms. Terms for (D) high and (E) low VitC sensitivity are ranked on $-\log(p \text{ value})$. (F), (G) Top 20 candidate markers associated with (F) high and (G) low VitC sensitivity ranked on log2FC. Gene ontology terms of known VitC related processes were further annotated. Significant proteins found in these VitC-related processes were also added.

Of note, prostate (Fig. S1), small cell lung (Fig. S2), lung adenocarcinoma (Fig. 5) and lung squamous cell (Fig. S3) cancer panels were analysed in the same manner. Interestingly, the prostate cancer panel behaved differently, with proliferation signals (e.g. cell cycle, telomere maintenance) being observed in the less sensitive cell lines, as opposed to in the more sensitive cell lines in all other cell line panels. Furthermore, lung cancer types and subtypes were overall quite sensitive but displayed different sensitivities: LUAD was the most sensitive panel (mean 0.42 ± 0.56 mM), followed by LUSC (mean 1.1 ± 0.62 mM) and SCLC (mean 1.8 ± 0.75 mM). In line with this, we found increased SLC2A3 (GLUT3) in LUAD sensitive cell lines (Fig. 5F), which is involved in the transmembrane transport of dehydroascorbic acid, the oxidized form of VitC. Iron regulators FLVCR1 and HMOX2 were additionally increased in this panel (Fig. 5F-G), which confirms the crucial role of iron and VitC uptake in VitC sensitivity.

3.3. Implications for combination treatment of chemotherapy-resistant lung cancer

Interestingly, among the top 5 proteins most enriched in highly VitC sensitive LUAD cancer cell lines were several candidates that have previously been implicated in chemotherapy resistance (TGM2(52) and VCP(53), Fig. 5F). Platinum-based chemotherapy is the central component of most treatment regimens administered to patients with NSCLC; however clinical benefit is often hampered by chemoresistance(54,55). Previously, we have profiled the lysates and secretomes of a panel of NSCLC cell lines in relation to cisplatin IC50 values, and revealed biofluid biomarker candidates with potential predictive value in determining cisplatin response(56). In the present study, we have extended the determination of cisplatin sensitivities to the entire LUAD panel. Strikingly, we found the least cisplatin sensitive cell line of the LUAD panel (H1975), as determined by assessing cell viability in response to different cisplatin doses, to be part of the more VitC sensitive cell line group. This further supports a role for VitC as a non-toxic alternative or combination treatment option for chemotherapy resistant lung cancer populations (Fig. 5A).

3.4. Pan-Cancer biology associated with VitC susceptibility

Our findings showed considerable heterogeneity at the level of individual proteins linked to VitC sensitivity per cancer cell panel, while strong common biology was shared between most of them. By studying the overlap between significantly enriched biological terms among cancer panels we found cell cycle, RNA metabolism, catalytic complex and biosynthesis to be associated with high VitC sensitivity (Fig. 6A). On the other hand, low VitC sensitivity was associated with extracellular vesicles/exosomes, mitochondrial terms, cell adhesion and protein transport terms across all panels (Fig. 6B).

By interrogating the relationship of the enriched hallmarks of cancer with VitC IC50s among panels, we observed 2 clusters of shared biology: adenocarcinomas PDAC, LUAD and CRC clustered away from PC, BC and SCLC (Fig. 6C). This difference could be influenced by hormone levels which are known to play a role in PC and BC, as well as in SCLC cell cultures(57). Overall, cancer cell lines with enriched proliferative hallmarks e.g. G2M checkpoint, mitotic spindle and Wnt beta catenin were more sensitive to VitC, whereas cell lines with strong survival signals like mTOR pathway, glycolysis, fatty acid metabolism were less sensitive, with Heme metabolism and Myc targets also enriched for a subset of panels. High proliferative signals were associated to high VitC sensitivity for all panels with the exception of the prostate cancer panel, which behaved in the opposite manner as observed earlier in the per panel analysis. Expression of proliferation marker MKI67 in more and less sensitive cell lines in different panels also showed this behaviour (Fig. S4A).

Merging all VitC associated proteins from the separate cancer cell panel analyses into one protein-protein interaction network, underscores the shared biology as can be evidenced by the high network connectivity (Fig. 7-8, Fig. S5-6). Pan-cancer network biology associated with high

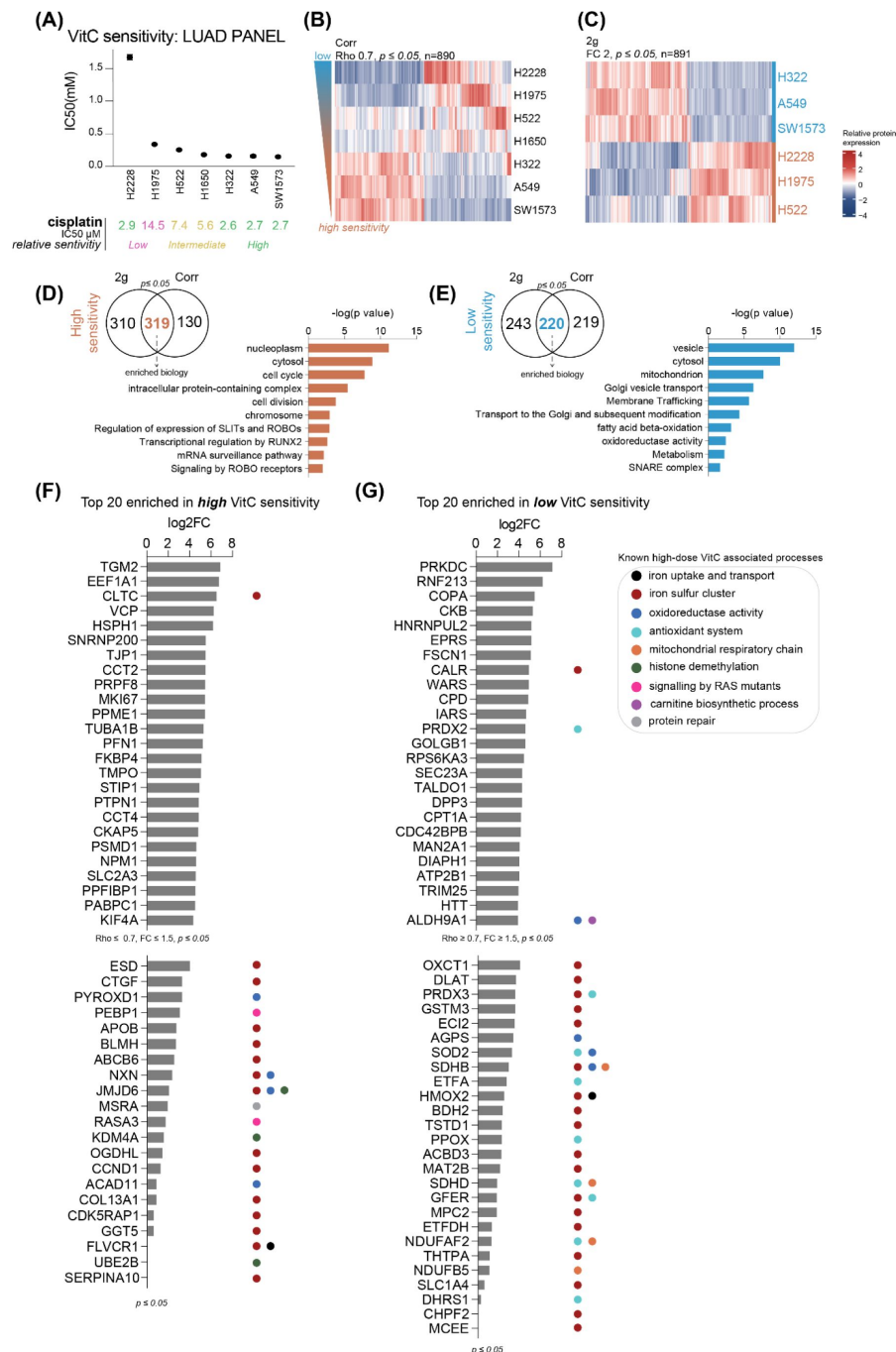


Fig. 5.

Lung adenocarcinoma VitC sensitivity and protein biomarker discovery analysis.

(A) Vitamin C sensitivities (IC50 mM) of $n=7$ LUAD cell lines. Cisplatin sensitivities (IC50s μM) were further annotated. (B) Hierarchical clustering heatmap of proteins correlated with high and low VitC sensitivity (Spearman Correlation $Rho \geq 0.7$, $pvalue \leq 0.05$, $n=890$ identified proteins). (C) Hierarchical clustering heatmap of proteins enriched in highly sensitive and lowly sensitive cell lines (2-group comparison, $FC \geq 2$, $pvalue \leq 0.05$, $n=891$ identified proteins). (D), (E) Venn diagram describing the overlap between significant proteins found in both statistical analysis, 2-group comparison and Spearman Correlation for (D) high and (E) low VitC sensitivity, as input for the top 10 enriched biology terms. Terms for (D) high and (E) low VitC sensitivity are ranked on $-\log_{10} pvalue$. (F), (G) Top 20 candidate markers associated with (F) high and (G) low VitC sensitivity ranked on $\log_2 FC$. Gene ontology terms of known VitC related processes were further annotated. Significant proteins found in these VitC-related processes were also added.

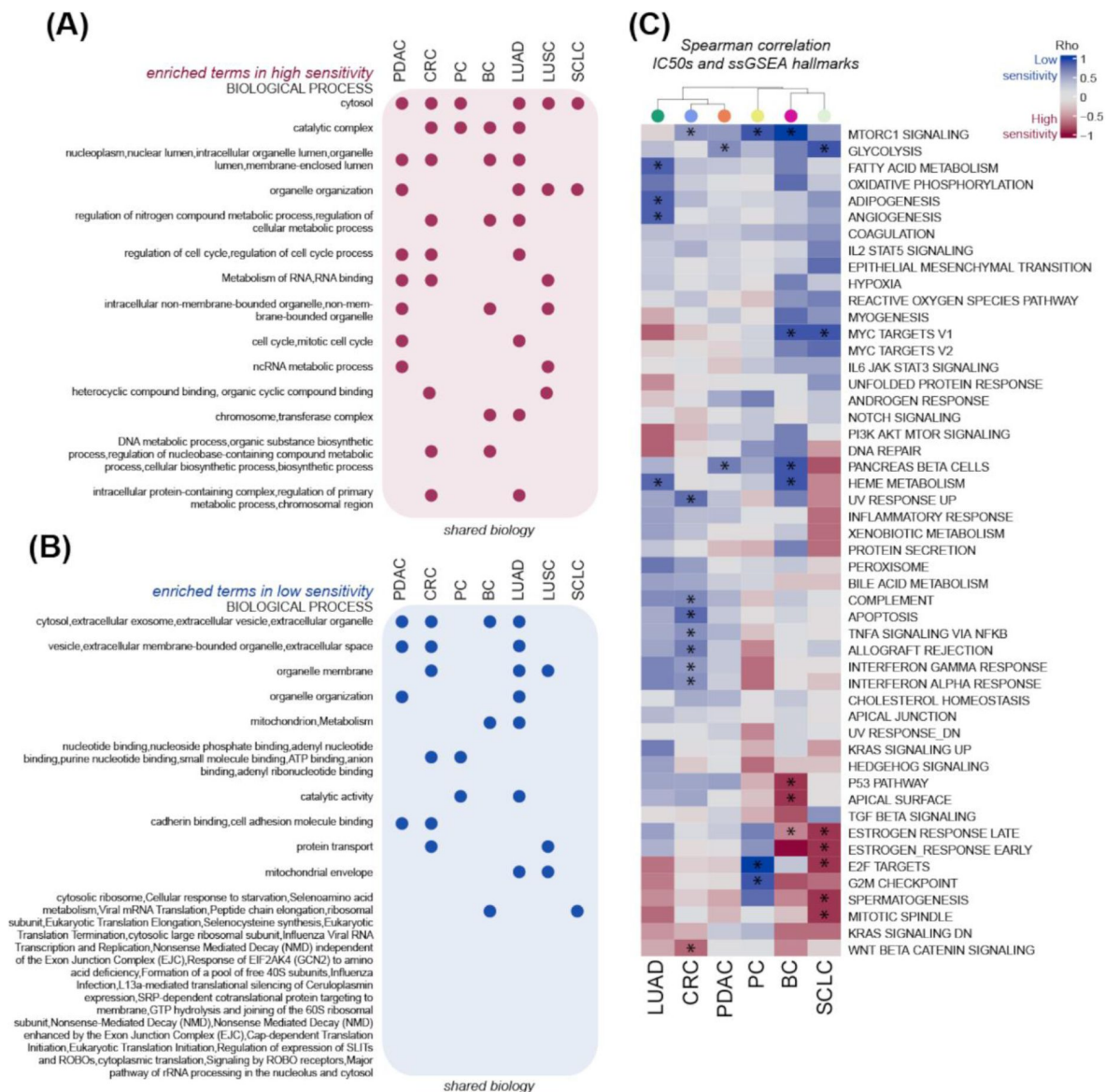


Fig. 6.

Enriched common biology associated with VitC sensitivity among cancer cell line panels.

Overlapping significant (*pvalue adjusted*) gProfiler terms (GO:BP, CC, MF and REACTOME) in (A) high sensitive cells and (B) less sensitive cells from shared proteins between spearman correlation and 2-group comparison analyses per panel. Maximum of top forty terms ordered by adjusted *pvalue* were used to assess term overlap between panels. (C) Cancer hallmarks (ssGSEA) associated with VitC sensitivity across cancer panels. Heatmap built with spearman correlation coefficients (Rho) between IC50s and ssGSEA hallmark enrichment scores per panel. Significant *pvalue* is annotated with an asterisk. LUSC panel was excluded from this analysis due to a low sample size (non-reliable results).

VitC sensitivity is shown in **Figs. 7** and S5 and low sensitivity in **Figs. 8** and S6. These analyses confirm and extend the gene ontology and hallmarks of cancer analysis with more functional details.

Overall, sensitive cancer cells exhibited higher expression of proteins involved in mRNA splicing and transcription, cell cycle, signal transduction and chromatin organization (**Fig. 7**). Another interesting protein cluster was involved in peroxisomal activity (ACAD11, ACOX1, SLC27A2, FADS2, EHHADH). Cancer cells with low sensitivity had elevated expression of proteins with functions in glycolysis and mitochondrial-related processes, vesicle-mediated transport, fatty-acid beta oxidation and Wnt receptor signalling pathway (**Fig. 8**). Importantly, we also identified two clusters with proteins involved in L-ascorbic acid binding and collagen fibril organization (PLOD2, COL3A1, P4HA1, LEPRE1) as well as Transferrin receptor activity linked to vesicle-mediated transport (TFRC, AP3S1, AP1B1, CPD) in the VitC low-sensitive group. Interestingly, a smaller vesicle-mediated transport cluster was found in sensitive cells, particularly expressed in the lung and pancreatic cancer panels. These proteins were involved in vesicle-membrane fusion (SNAP23, VTI1A, YKT6, STX8, STXBP1), exocytosis and secretion (RAB3A, STXBP6), as well as in insulin-dependent trafficking of GLUT4, which can be increased on the cell surface and enhance glucose transport (TBC1D4). Vesicle-mediated transport cluster identified in less sensitive cells showed a clearly distinct function, e.g. COPI (COPA, COPE), COPII (SEC24A, SEC24B, SEC23A, SEC23IP), and TRAPP (TRAPPC4, TRAPPC6B) complexes, and SNAP receptor activity and SNARE binding activity (NAPA, BET1L, STX12, STX6, STX10, VAMP3). Interestingly, retrograde golgi-to-ER transport proteins (ARCN1, STX8, SCFD1) and glycosylation proteins (LMAN2, GOLGA2) were also identified.

Among the proteins shared in more than two or three panels (**Fig. 7A**, **Fig. 8A**), two proteins, HCCS and PDCD4, were found to be significantly associated with low sensitivity in 3 cancer cell panels (Fig. S4B). Interestingly, HCCS is a lyase that catalyses the transfer of a mitochondrial heme group for the functioning of cytochrome c (58). Given the importance of accessible free iron (2), we propose that elevated cytochrome c levels could suggest diminished levels of available iron, thereby reducing the pro-oxidant effect of high dose VitC. Besides this, decreased tumour suppressor PDCD4, found in 3 cancer cell panels with high VitC sensitivity (Fig. S4B), has been linked to increased proliferation in glioma cells (59), which supports our findings. In addition, other multiple-panel proteins associated with reduced VitC sensitivity were involved in vesicle-mediated transport (VAMP3), fatty acid oxidation (BDH2, ETFA), ERK signalling (RPS6KA3), proteasome (PSME2), calcium homeostasis (S100A6) and mitochondrial homeostasis and respiration (HSPA9, COQ6) (**Fig. 8**, Fig. S6, see protein names highlighted in bold). Two pan-cancer proteins associated with high VitC sensitivity were the microtubule dynamics and mitosis regulator CKAP5 (60, 61), and MORF4L2, implicated in chromatin assembly and histone modification (Fig. S4B). Other proteins shared between sensitive cell lines from multiple cancer panels were also found to be involved in cell cycle (NCAPG, LIG1, PDS5A, TMPO, PPP1CB, PPME1, CAPRIN1, PKP4, SYNE2), RNA splicing (SF3B6, DDX46, SNRNP200, HCFC1, THOC5, PRPF38B) and chromatin organization (RBBP7, HCFC1) (**Fig. 7**, Fig. S5).

3.5. (Phospho-)proteomic changes upon high-dose VitC treatment

To assess the effect of high dose vitC on the proteome and phosphoproteome in a tumour type with poor prognosis, a subset of the PDAC panel, displaying a broad range of IC₅₀ values, was analysed (**Fig. 1** and **2**). Given the poor prognosis of this tumour type, we aimed to further elucidate the mechanism of action by examining on-treatment samples, thereby gaining a deeper understanding on tumour cell responses to VitC. First, we confirmed an increase in intracellular ROS levels with time on treatment (**Fig. 9A-B**), and a complete recovery of cell growth was achieved upon the addition of catalase, a well-known hydrogen peroxide scavenger (**Fig. 9C-D**). Subsequently, we performed phospho-proteomic profiling of four PDAC cell lines treated with VitC at their respective IC₅₀ values at three distinct time points (2 hours, 4 hours, and 24 hours, **Fig. 9E**). This approach allowed us to capture early changes in the phospho-proteome as well as expression changes in the proteome. From the 38 samples, 3 samples were excluded from further

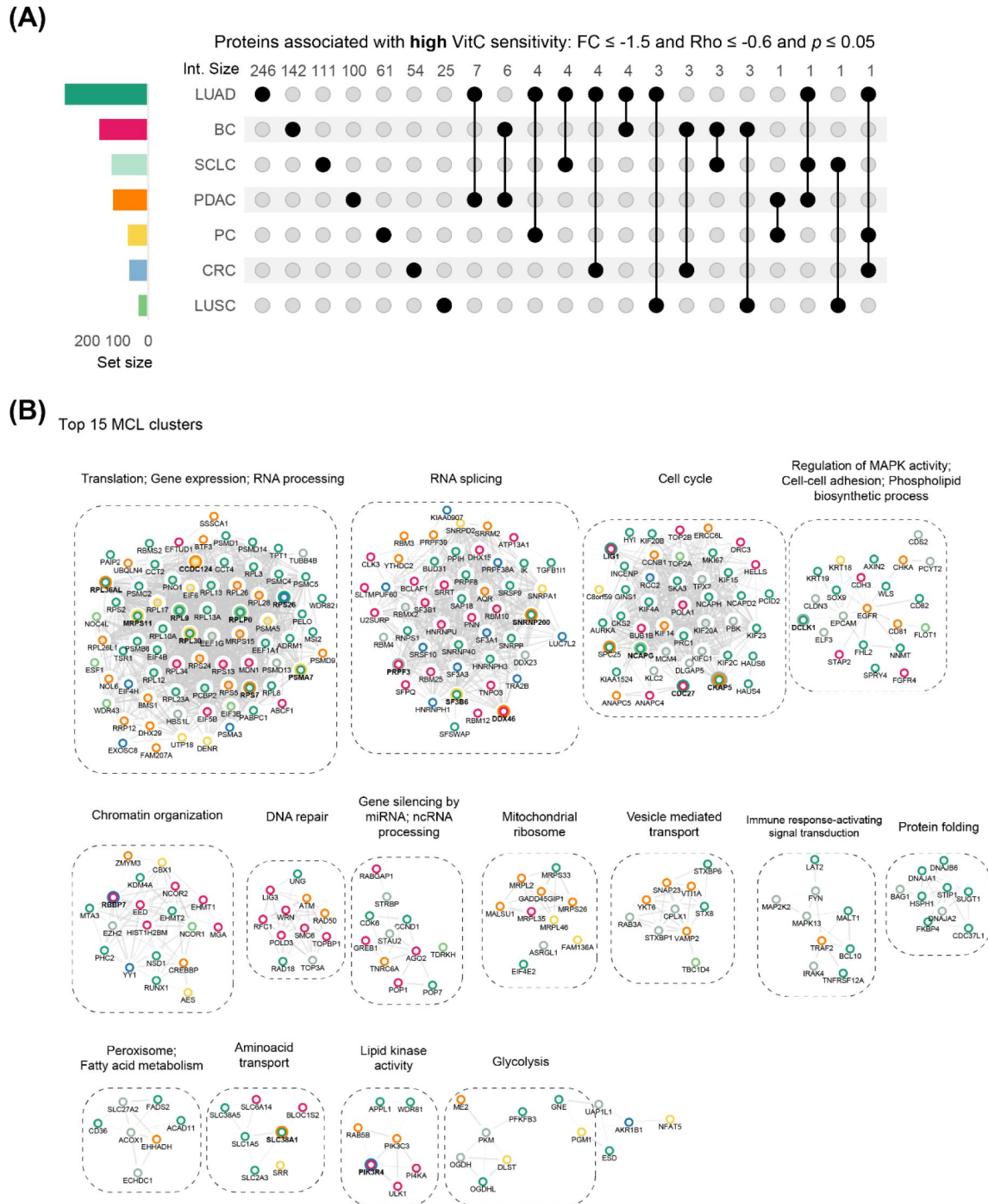


Fig. 7.

Pan-cancer biology associated with high VitC sensitivity.

(A) Upset plot highlighting the overlap of proteins associated with high VitC sensitivity identified in each cancer panel. Overlap between panels is indicated with joined black circles. (B) Top 15 protein-protein interaction clusters and enriched biology. All proteins in panel A were used as input for the networks. Node colours refer to cell line panel, see [Fig. 7A](#). Bold nodes refer to shared proteins between at least two cancer cell panels. See the expanded network in Fig. S5.

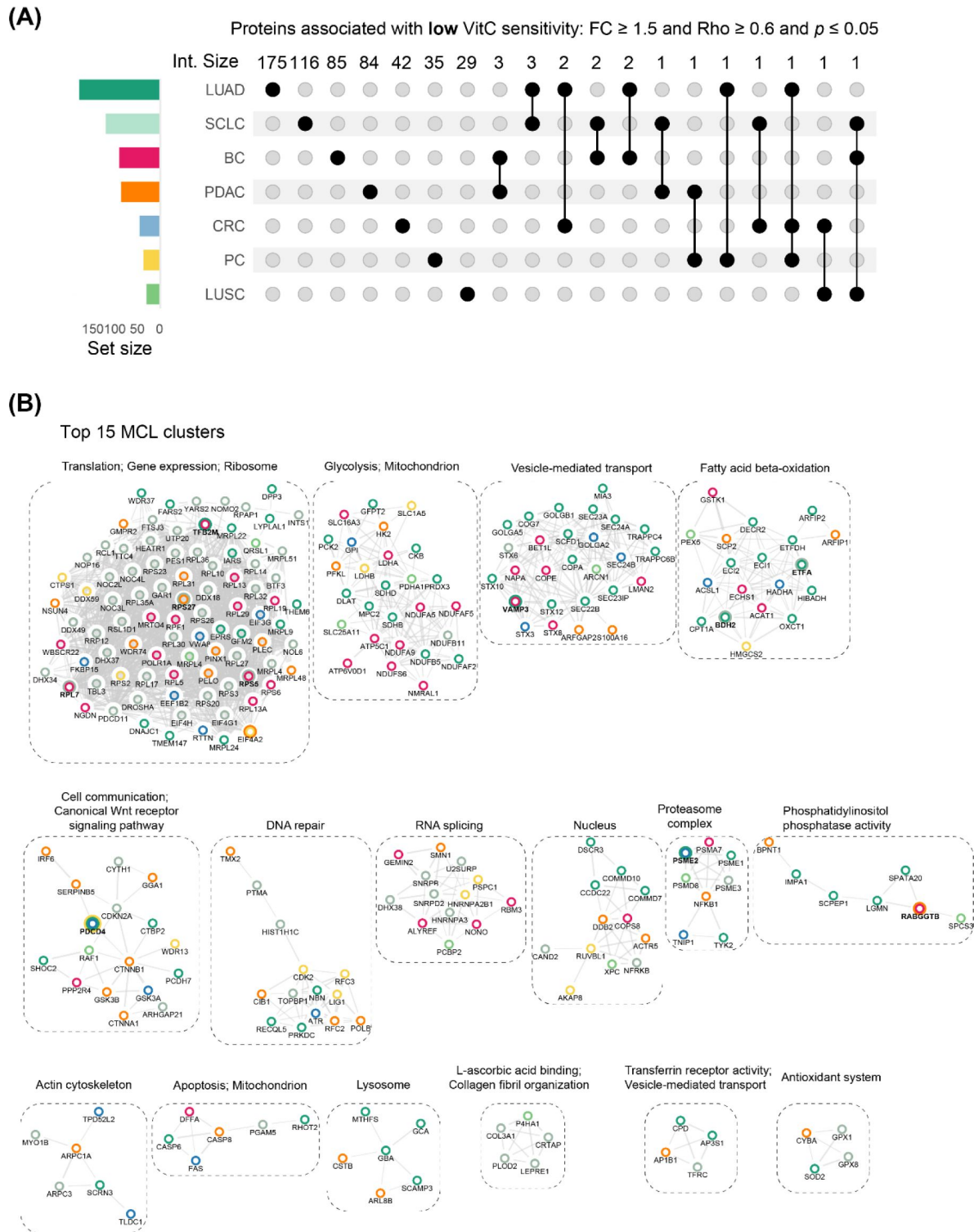


Fig. 8.

Pan-cancer biology associated with low VitC sensitivity.

(A) Upset plot highlighting the overlap of proteins associated with low VitC sensitivity identified in each cancer panel. Overlap between panels is indicated with joined black circles. (B) Top 15 protein-protein interaction clusters and enriched biology. All proteins in panel A were used as input for the networks. Node colours refer to cell line panel, see [Fig. 8A](#). Bold nodes refer to shared proteins between at least two cancer cell panels. See the expanded network in Fig. S6.

analyses due to the low amount of phosphopeptides measured compared to replicates and external control sample HCT116 (Fig. S7A,B). Thus, a total of 35 samples, with an average of 3296 proteins and 7617 phosphosites per sample (Fig. S7C) were subsequently used in unsupervised clustering (Fig. S7D,E). By performing an unsupervised clustering, we observed that data primarily clustered based on the cell line type, and to a lower extent by treatment time point (Fig. S7D,E). Therefore, we followed a cell line-specific approach for data analysis. This approach allowed us to identify and compare cell line-specific mechanisms in all cell lines in order to uncover both unique and common changes in the (phospho-) proteome induced by VitC treatment.

By examining proteomic changes and cancer hallmarks, immunosuppressive and survival pathways were decreased upon high-dose VitC. IFN- α response was particularly enhanced upon VitC. Interestingly, IFN- α response can be activated in tumour cells upon cell damage, which can trigger activation of macrophages, NK cells, and T cells, and eventually promotes cancer cell killing. This is in line with *in vivo* studies reporting increased IFN- α upon VitC anti-viral immune responses(62), as well as increased IFN- γ upon the synergistic combination of VitC coupled to immune checkpoint blockade(21). Similarly, upon 24h of VitC treatment, the majority of cell lines showed a decrease in TGF- β and IL6 JAK-STAT3 signalling, two key immunosuppressive pathways in PDAC tumours that can positively regulate and attract myeloid-derived suppressor cells and regulatory T cells to the tumour microenvironment(63). Furthermore, Wnt/ β -catenin signalling, which also holds an immunosuppressive role in PDAC tumours(64), was found to be reduced upon VitC, in line with previous studies in PDAC(65).

By interrogating individual proteins, we observed only a few significant changes in the proteome after the relatively early time point of 24h. This suggests that longer treatment duration (~48h) may have been more suitable to capture significant alterations in the proteome. Despite this, we were able to identify interesting regulations, such as the decrease of iron receptor TFRC and IMMT expression (maintains the mitochondrial cristae morphology), or the increase of CYCS expression, which may be an indicator of mitochondrial outer membrane permeabilization(66) (Fig. S7F). Overall, proteome changes suggested that 24h after high-dose VitC treatment, the majority of PDAC cell lines displayed reduced immunosuppressive and survival signals, and overall increased iron metabolism, in line with reported mechanisms in multiple cancer types(1).

Most identified phosphosite changes occurred on the cell line treated with the highest VitC IC50 dose, i.e. Hs 766T with 2mM. In particular, we observed a significant regulatory impact on various phosphorylation events on proteins involved in cell cycle arrest (RB1, AHNAK), cell cycle progression (e.g. GRK5, CCDC86), cell adhesion and invasion (e.g. CTNND1, PALLD, ITGB4), signal transduction (e.g. PTPRA, SPAG9) and endocytosis (e.g. FAM21, SDPR). Phosphorylation site changes of particular significance were identified in proteins including GLUT1/SLC2A1, RB1, MARCKS, PTPRA, and CTNND1. Specifically, the phosphorylation site S266 of GLUT1 exhibited an upregulation at 2 hours and 4 hours, followed by a downregulation at 24 hours. Notably, this phosphorylation event is known to enhance the translocation of GLUT1 to the plasma membrane, facilitating glucose uptake(67). The fact that GLUT1 phosphorylation was upregulated in the early hours of treatment may suggest that oxidized VitC (DHA) and glucose are competing for cell uptake. Thus, its subsequent downregulation at 24h could be a cell-intrinsic mechanism to reduce the VitC uptake. Likewise, phosphorylation of FAM21A S802/S1114 and FAM21C S1052, both proteins involved in recycling GLUT1(68), were found to be upregulated over time. This upregulation holds significance, considering that GLUT1 is essential for transporting DHA into the cells.

For the cells treated with lower doses, we also observed a general increase in phosphorylation occurring in cell cycle arrest proteins after 24h (e.g. AHNAK), as well as in cell adhesion markers (e.g. LMNA, FLNB, FMNL2) and splicing regulators (e.g. SRRM2) (data not shown). Interestingly, phosphorylation on mTORC2 subunit RICTOR-T1332, was downregulated in Capan-2, and negative mTOR-regulatory AKT1S1-S203 was upregulated in Suit-2.

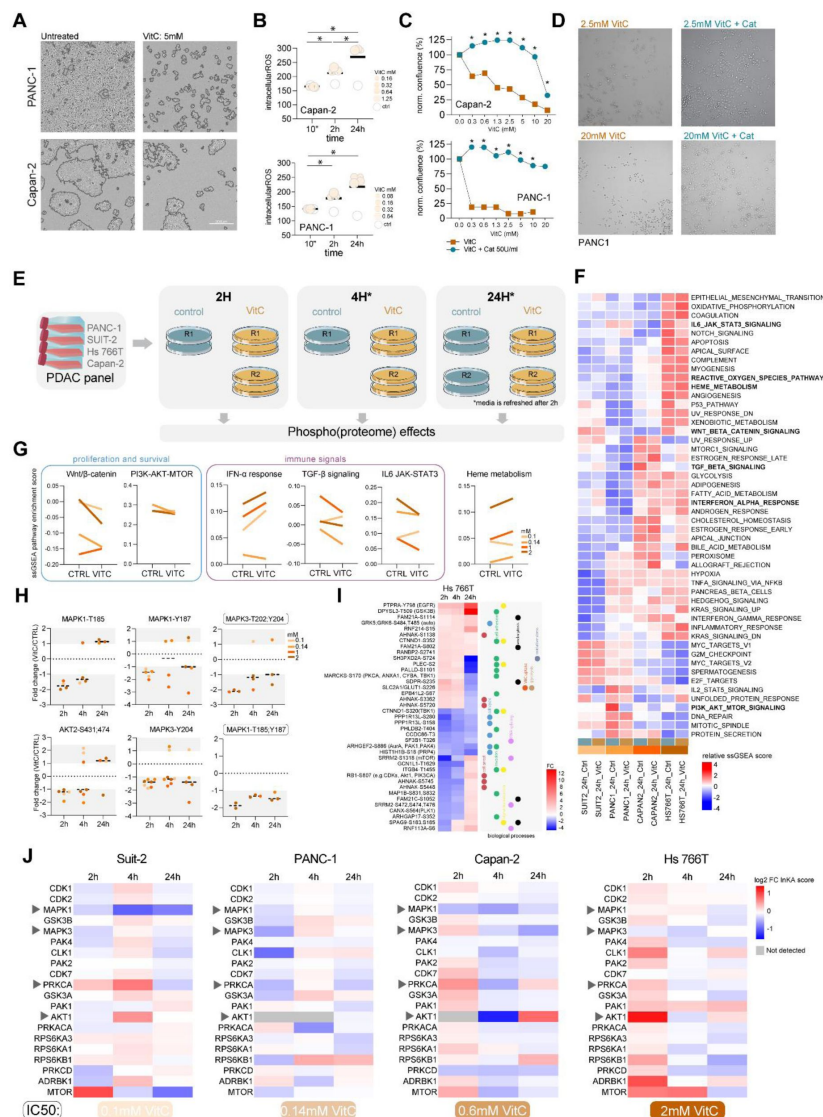


Fig. 9.

Effects of VitC on the proteome and phospho-proteome of a subset of PDAC cells.

(A) Cell confluence images of PANC-1 and Capan-2 cell lines for treated (5mM) and untreated conditions. (B) Intracellular ROS levels upon multiple VitC concentrations in PANC-1 and Capan-2 cell lines after 10 minutes, 2h and 24h. For every time point, intracellular ROS levels are normalized to the levels of the untreated condition. Data from n=2 biological replicates. Statistical significance between timepoints was calculated by a paired t-test. (C) Cell growth of PANC-1 and Capan-2 upon VitC, and upon VitC followed by the exposure to catalase. (D) Representative images of PANC-1 cells VitC treated (2.5 and 20mM) with and without catalase exposure. (E) Workflow scheme of phospho-proteomic profiling of VitC treated cell lines at 2h, 4h and 24h. See Methods. (F) Unsupervised clustering based on ssGSEA results using hallmarks of cancer. Enrichment scores shown as z-scores. (G) Interesting ssGSEA enrichments between control and VitC samples after 24h. Colours depict cell lines in order of VitC sensitivity and VitC dose (0.1mM Suit-2, 0.14mM PANC-1, 0.6mM Capan-2, 2mM Hs 766T). (H) Phosphosite changes upon VitC at 2h, 4h and 24h, calculated by FC between control and VitC, of downstream regulators MAPK1/3 and AKT2 in all cell lines. Colours depict cell lines in order of VitC sensitivity and VitC dose (0.1mM Suit-2, 0.14mM PANC-1, 0.6mM Capan-2, 2mM Hs 766T). Rectangle indicates double phosphorylation required for MAPK1/3 activation. (I) Phosphosite changes upon VitC at 2h, 4h and 24h, calculated by FC between control and VitC in Hs 766T cell line. (J) Kinase activity changes upon VitC treatment at 2h, 4h and 24h timepoints. Log2 fold changes were calculated for the top 20 ranked kinases based on the cumulative INKA score. A fold change cut-off of (-) 1.5 was used to determine which kinases are altered upon VitC treatment. In grey, FC values cannot be determined due to the absence of detected INKA kinases in at least one of the replicates, either in the VitC or control group. Arrows indicate members of MAPK and PI3K/AKT signalling.

By looking at relevant downstream regulators of KRAS mutant tumours, e.g. MAPK and PI3K/AKT pathways, we observed distinct effects among cell lines, with a regulatory decreasing trend for the majority of them (**Fig. 9H**). The intensities of both phosphosites MAPK3 T202/Y204 and MAPK1 T185/Y187, which are important for ERK pathway activation, were decreased upon VitC, particularly 2h after treatment. Single phosphorylations MAPK1-Y187, MAPK1-Y185 and MAPK3-Y204, showed similar trends. AKT1 phosphosites were not identified in our set up, however we detected AKT2 S431/474, which showed decreased phosphorylation at 2 and 4 hours for PANC-1, while Capan-2 and Hs 766T remained nearly unchanged. Suit-2, on the contrary, showed an increase 4 hours after treatment. Interestingly, AKT2-S474 is known to be crucial for insulin-stimulated mTOR activation and glucose uptake in adipocytes(69).

Kinase activity changes were then evaluated upon VitC overtime in the top 20 rankings by using the INKA pipeline. Of note, differential changes in kinase activity were observed in three out of four cell lines, including MAPK1, AKT1, and mTOR kinases (**Fig. 9J**). For instance, MAPK1/3 kinase activity decreased consistently across all time points, particularly in Suit-2 and Capan-2, with a decrease at 24h for most of the cell lines. Additionally, AKT1 showed a downregulation in Capan-2 at 4 hours, followed by an upregulation at 24 hours. For Hs 766T, upon VitC, AKT1 returned to control levels after 24h, whereas mTOR and RPS6KB1 decreased over time. The most consistent change was mTOR kinase activity, which declined over time in most cell lines, consistent with the reduced activity of the PI3K-AKT-MTOR pathway at the protein level (**Fig. 9G**). These findings are in line with previous studies in thyroid(15) and breast(70) cancer cells showing inhibition of PI3K-AKT-MTOR and MAPK pathways upon high-dose VitC, further supporting our results.

Although not in the top 20 rankings, kinase activity of CAMK2D and CAMK2A, involved in antioxidant mechanisms by sensing ROS-mediated stress levels(71), were downregulated upon VitC treatment, particularly after 24h, in both Capan-2 and PANC-1 cell lines (data not shown). This data may suggest that antioxidant mechanisms are overcome after 24h, in line with prooxidant-mediated cytotoxicity (**Fig. 9A,B**).

In addition, phospho-proteomics of VitC IC50-treated cell lines revealed cell line specific responses. For instance, in the most sensitive cell line cell migration, epithelial-mesenchymal transition (EMT) and cell growth were decreased in early response while the least sensitive cell line showed increased GLUT1 translocation, cell cycle arrest and oxidative stress.

4. DISCUSSION

In high concentrations (mM range), which can only be reached by intravenous administration, VitC has proved to exert potent anti-cancer effects in multiple cancer types. Here we report VitC IC50 values in relation to proteome features and associated biological functions for a large pan-cancer cell panel (n=51) representing major solid cancer types.

Importantly, all cancer cell lines tested in this study were particularly sensitive, with an average IC50 of 1.7 ± 0.4 mM, with only 5 out of 51 cell lines requiring a 5-10 mM dose. Overall, all tested IC50 values were within the clinically achievable range, in line with previous studies(1,2,72). Furthermore, our results for PDAC, CRC, BC and LC also showed that VitC sensitivity is independent of cancer subtypes (e.g. epithelial/mesenchymal or CMS subtypes), which positions VitC as a promising one-size-fits-all anti-cancer agent. The baseline proteome data showed that high VitC sensitivity is associated with proliferation-related functions, while lower sensitivity is linked to metabolic processes and vesicle-mediated transport. Finally, (phospho-)proteome analysis of on-treatment vit C effects revealed down-modulation of MAPK1/2, AKT-MTOR signalling and immune suppressive signalling, while IFN- α response was enhanced upon VitC.

4.1. Confirmation of well-established VitC effects

By exploring known proteins associated with VitC susceptibility in cancer, we were able to confirm previous insights on the importance of iron metabolism, antioxidant system, ROS-dependent pathways, metabolic processes and DNA damage proteins. In particular, high doses of VitC are known to utilize intracellular iron levels to increase ROS levels. As a result, cancer cells with elevated free labile iron levels and low catalase are generally more susceptible to the pro-oxidant VitC cytotoxic effect. By interrogating known proteins involved in iron metabolism, we found iron transport members like DNMT1, ZIP14 and FLVCR1 were found to be increased in more sensitive cell lines, which underscores the role of iron trafficking and the intracellular free iron pool as the key player in VitC sensitivity. These members are involved in non-TF-bound iron pathway, typically occurring during iron overload(73 [↗](#)). Remarkably, iron importer TFRC was found to be enriched in less sensitive cell lines. We hypothesize that balanced expression of these members is crucial for increased labile iron pool levels, which are crucial to promoting the VitC pro-oxidant effect in cancer cells, as previously described(2 [↗](#)).

4.2. Proliferative functions

Findings from both individual cancer panels and our comprehensive pan-cancer panel analyses indicate an association between proliferative functions and high sensitivity to VitC. Notably, proliferative signals like elevated EGFR and IGF2R were also linked to increased VitC sensitivity in the PDAC panel. In line with this, we identified four potential pan-cancer proteins highly associated with VitC sensitivity, with mitotic, chromatin and histone regulators CKAP5 and MORF4L2 specifically found to be increased in highly sensitive cell lines in SCLC, LUAD and PDAC, and LUAD, CRC and PC panels, respectively. Interestingly, a pan-cancer protein found to be reduced in sensitive cell lines in BC, LUSC and SCLC panels was PDCD4, a tumour suppressor found to be reduced in highly proliferative glioma cells(59 [↗](#)), consistent with our findings.

4.3. Mitochondrial functions, iron and oxidative stress levels

Cancer cells with VitC IC50 values in the mM range exhibited, as expected, higher abundance of antioxidant defence proteins, as well as higher levels of proteins with vesicle transport and mitochondrial functions. For instance, SPR protein was found to be associated with low VitC sensitivity in the CRC panel. This protein is a promoter of BH4, which is a potent antioxidant that may be involved in ferroptosis induction(45 [↗](#)). Intriguingly, BH4 levels and its antioxidant activity can regulate iron metabolism and mitochondrial function of T cells too. From the pan-cancer network analysis, multiple antioxidants such as CYBA, GPX1, SOD2 and GPX8 were also identified in less sensitive cell lines. In the BC panel, GSTK1, which is an antioxidant contained in peroxisomes, correlated with low VitC sensitivity, and has been found to be involved in ferroptosis together with CAT, SOD1 and PRDX5(74 [↗](#)). This cancer cell death process has been linked to VitC anti-cancer effect in several recent studies and may be further studied(75 [↗](#), 76 [↗](#)).

These metabolic and mitochondrial functions were confirmed in the common pan-cancer analysis, which supports that energy production processes can also be targeted by VitC(77 [↗](#)). Wnt beta catenin signalling significantly correlated with high VitC sensitivity in the CRC panel, however from the network analysis we also identified members of this pathway (e.g. CTNNA1, CTNNB1, GSK3A/B) correlating with low sensitivity in other panels. To some extent, beta catenin could be involved in VitC sensitivity in cancer cells since it plays a key role in heme synthesis, glucose uptake and fatty acid metabolism, and positively regulates L-gulonolactone oxidase, crucial for VitC biosynthesis(78 [↗](#)). Furthermore, VitC has been recently shown to inhibit EMT in PDAC via the Wnt/beta catenin pathway(65 [↗](#)), which underscores VitC's multi-targeting capability in cancer. Besides, mTOR was the pathway most significantly associated with reduced VitC sensitivity. This pathway is key for cancer survival, and it has recently been shown to be inhibited by VitC in vivo(79 [↗](#)).

A pan-cancer protein associated with lower VitC sensitivity was HCCS, which catalyses the heme group linking, crucial for cytochrome c synthesis. We suggest that mature cytochrome c has the ability to associate with free iron, consequently mitigating oxidative stress induced by high doses of vitamin C. High-dose VitC is known to reduce cytochrome c levels, leading to impaired mitochondrial respiration, reducing ATP levels and yielding superoxide(80 [↗](#)). However, cancer cells are able to surpass mitochondrial ATP production by promoting aerobic glycolysis, known as the Warburg effect. Our findings reveal an enrichment of this process in cancer cells exhibiting diminished sensitivity to VitC. These mechanisms, targeting the energy levels and increasing reactive oxygen species, together with sensitivity associated proliferative signals, make high-dose VitC a very promising candidate for anti-cancer therapy comparable to chemotherapeutics or mitochondrial targeting agents, yet with reduced side effects.

Furthermore, our findings position VitC as a potential treatment for chemoresistant tumours. In the LUAD panel, the cell line resistant to cisplatin was highly sensitive to vitC. Interestingly, PDAC's top candidate associated with higher VitC sensitivity was ANPEP, a protein associated with multi-drug resistance, also to chemotherapeutic drugs, in multiple cancer types(81 [↗](#)). Other proteins involved in chemoresistance, namely TGM2 and VCP, were associated with high VitC sensitivity in the LUAD panels. When considering our evaluation of cisplatin responses across the LUAD cell line panel, this further underscores VitC as a potential alternative treatment for chemoresistant tumours.

Altogether, these findings position high-dose VitC as a broad anti-cancer agent comparable to chemotherapeutic or mitochondrial disruptor agents. Particularly for heavily treated patients with hard-to-treat tumours such as PDAC, CRC or NSCLC, a vast amount of preclinical evidence supports high-dose intravenous VitC as a promising alternative(1 [↗](#)). Promising phase I/II clinical studies are continuously showing VitC as a tolerable, safe and effective anti-cancer agent in combination with standard treatments, e.g. in NSCLC and GBM(2 [↗](#)) and in PDAC(82 [↗](#)). Larger scale randomized phase III clinical studies are warranted to support its clinical efficacy.

4.4. On-treatment VitC effects in PDAC cell lines

High dose VitC at IC50 values induced both cell-specific alterations as well as common down-modulation of cell survival signalling and differential effects on immune signalling, shifting cells to less immune suppression. In contrast to the high number of phosphosite alterations after 2-4h, we noted only a limited number of changes in protein expression following 24 hours of VitC treatment. The relatively brief duration for detecting protein expression changes, coupled with the use of lower doses (< 1 mM), may explain why we did not see changes in EMT hallmark proteins, e-cadherin or vimentin, contrary to what was previously described in PDAC cell lines(83 [↗](#)). Possibly, more protein expression changes may be found by including later timepoints (>48h), in particular those related to tumour survival and growth.

5. CONCLUSION

Our findings showed remarkable VitC sensitivity of all cell lines within the clinically achievable range, suggesting that predictive markers are unnecessary. The pro-oxidant effects and high proliferation signature observed in highly sensitive cell lines provides valuable support for positioning VitC as an alternative agent to chemotherapy, which is also shown to specifically target cancer cells rather than normal cells(2 [↗](#), 72 [↗](#)). Moreover, the significant increase in antioxidant and mitochondrial activity observed in less sensitive cell lines reinforces the well-established mechanism of action of VitC. Finally, the on-treatment analysis in a subset of the PDAC panel, revealed changes in the phospho-proteome and kinase activity involving proliferation and survival, invasion and immune signals. The findings from this study underscore VitC's

effectiveness against multiple cancer types, independent of subtypes, thereby laying the foundation for positioning this compound as a promising anti-cancer agent that warrants widespread clinical studies, as well as use in the palliative setting.

Abbreviations

- VitC: vitamin C
- PDAC: pancreatic ductal adenocarcinoma
- CRC: colorectal cancer
- LUAD: lung adenocarcinoma
- LUSC: lung squamous cell carcinoma
- SCLC: small cell lung cancer
- PC: prostate cancer
- BC: breast cancer
- ROS: reactive oxygen species

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Conflict of interest

The authors declare that they have no competing interests.

Author contributions

A.V.M. and F.B. designed the study, performed experiments, analysed data, performed bioinformatic analyses, and wrote the manuscript. K.T., E.H.Y.Y., L.M., S.S. and D.D. were involved in cell culture and VitC sensitivity experiments. M. M., T.L.L., I.B., A.E. and T.S. were involved in proteomic sample processing. S.R.P. performed the mass spectrometry measurements and S.R.P., T.V.P. and J.C.K. analysed data. E.G. and J.P.M. collaborated by providing cell lines. C.R.J. designed and supervised the study and wrote the manuscript.

Peer review

Data accessibility

The mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium via the PRIDE(84⁸⁴) partner repository with the dataset identifier PXD046089. Human Swiss-Prot database used for raw data search was acquired from the UniProt database (<https://www.uniprot.org/>).

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References

1. Böttger F, Vallés-Martí A, Cahn L, Jimenez CR (2021) **High-dose intravenous vitamin C, a promising multi-targeting agent in the treatment of cancer** *J Exp Clin Cancer Res* **40**
2. Schoenfeld JD, Sibenaller ZA, Mapuskar KA, Wagner BA, Cramer-Morales KL, Furqan M, et al. (2017) **O₂⁻ and H₂O₂-Mediated Disruption of Fe Metabolism Causes the Differential Susceptibility of NSCLC and GBM Cancer Cells to Pharmacological Ascorbate** *Cancer Cell* **31**:487–500
3. Ngo B, Van Riper JM, Cantley LC, Yun J (2019) **Targeting cancer vulnerabilities with high-dose vitamin C** *Nat Rev Cancer* **1**
4. Reczek CR, Chandel NS (2015) **Revisiting vitamin C and cancer** *Science (80-)* **350**:1317–8
5. van der Reest J, Gottlieb E (2016) **Anti-cancer effects of vitamin C revisited** *Cell Res* **26**:269–70
6. Du J, Cullen JJ, Buettner GR (2012) **Ascorbic acid: Chemistry, biology and the treatment of cancer** *Biochim Biophys Acta* **1826**
7. Michels AJ, Frei B (2013) **Myths, artifacts, and fatal flaws: Identifying limitations and opportunities in vitamin C research** *Nutrients* **5**:5161–92
8. Hayes JD, Dinkova-Kostova AT, Tew KD (2020) **Oxidative Stress in Cancer** *Cancer Cell* **38**:167–97
9. Sullivan LB, Chandel NS (2014) **Mitochondrial reactive oxygen species and cancer** *Cancer Metab* **2**
10. Porporato PE, Filigheddu N, Pedro JMBS, Kroemer G, Galluzzi L (2017) **Mitochondrial metabolism and cancer** *Cell Res* **28**:265–80
11. Doskey CM, Buranasudja V, Wagner BA, Wilkes JG, Du J, Cullen JJ, et al. (2016) **Tumor cells have decreased ability to metabolize H₂O₂: Implications for pharmacological ascorbate in cancer therapy** *Redox Biol* **10**:274–84
12. Zeng LH, Wang QM, Feng LY, Ke YD, Xu QZ, Wei AY, et al. (2019) **High-dose vitamin C suppresses the invasion and metastasis of breast cancer cells via inhibiting epithelial-mesenchymal transition** *Onco Targets Ther* **12**:7405–13
13. Zhao L, Quan Y, Wang J, Wang F, Zheng Y, Zhou A (2015) **Vitamin C inhibit the proliferation, migration and epithelial-mesenchymal-transition of lens epithelial cells by destabilizing HIF-1 α** *Int J Clin Exp Med*
14. Ma Y, Chapman J, Levine M, Polireddy K, Drisko J, Chen Q (2014) **High-Dose Parenteral Ascorbate Enhanced Chemosensitivity of Ovarian Cancer and Reduced Toxicity of Chemotherapy** *Sci Transl Med* **6**:222–222
15. Su X, Shen Z, Yang Q, Sui F, Pu J, Ma J, et al. (2019) **Vitamin C kills thyroid cancer cells through ROS-dependent inhibition of MAPK/ERK and PI3K/AKT pathways via distinct mechanisms** *Theranostics*

16. Su X, Li P, Han B, Jia H, Liang Q, Wang H, et al. (2021) **Vitamin C sensitizes BRAFV600E thyroid cancer to PLX4032 via inhibiting the feedback activation of MAPK/ERK signal by PLX4032** *J Exp Clin Cancer Res* **40**
17. Lee KE, Hahm E, Bae S, Kang JS, Lee WJ (2017) **The enhanced tumor inhibitory effects of gefitinib and L-ascorbic acid combination therapy in non-small cell lung cancer cells** *Oncol Lett* **14**:276–82
18. Wang L, Luo X, Li C, Huang Y, Xu P, Lloyd-Davies LH, et al. (2017) **Triethylenetetramine Synergizes with Pharmacologic Ascorbic Acid in Hydrogen Peroxide Mediated Selective Toxicity to Breast Cancer Cell** *Oxid Med Cell Longev*
19. Yang G, Yan Y, Ma Y, Yang Y (2017) **Vitamin C at high concentrations induces cytotoxicity in malignant melanoma but promotes tumor growth at low concentrations** *Mol Carcinog*
20. Sinha BK, van 't Erve TJ, Kumar A, Bortner CD, Motten AG, Mason RP. (2017) **Synergistic enhancement of topotecan-induced cell death by ascorbic acid in human breast MCF-7 tumor cells** *Free Radic Biol Med*
21. Magri A, Germano G, Lorenzato A, Lamba S, Chilà R, Montone M, et al. (2020) **High-dose vitamin C enhances cancer immunotherapy** *Sci Transl Med* **12**
22. Mansoor F, Kumar S, Rai P, Anees F, Kaur N, Devi A, et al. (2021) **Impact of Intravenous Vitamin C Administration in Reducing Severity of Symptoms in Breast Cancer Patients During Treatment** *Cureus* **13**
23. Günes-Bayir A, Kiziltan HS (2015) **Palliative Vitamin C Application in Patients with Radiotherapy-Resistant Bone Metastases: A Retrospective Study** *Nutr Cancer*
24. Aumailley L, Bourassa S, Gotti C, Droit A, Lebel M (2021) **Vitamin C Differentially Impacts the Serum Proteome Profile in Female and Male Mice** *J Proteome Res* **20**:5036–53
25. Vichai V, Kirtikara K (2006) **Sulforhodamine B colorimetric assay for cytotoxicity screening** *Nat Protoc* **1**:1112–6
26. Keepers YP, Pizao PE, Peters GJ, van Ark-Otte J, Winograd B, Pinedo HM (1991) **Comparison of the sulforhodamine B protein and tetrazolium (MTT) assays for in vitro chemosensitivity testing** *Eur J Cancer Clin Oncol* **27**:897–900
27. Buranasudja V, Doskey CM, Gibson AR, Wagner BA, Du J, Gordon DJ, et al. (2019) **Pharmacologic ascorbate primes pancreatic cancer cells for death by rewiring cellular energetics and inducing DNA damage** *Mol Cancer Res* **17**:2102–14
28. Vallés-Martí A, Mantini G, Manoukian P, Waasdorp C, Sarasqueta AF, Haas RR de G, et al. (2023) **Phosphoproteomics guides effective low-dose drug combinations against pancreatic ductal adenocarcinoma** *Cell Rep* **42**
29. Piersma SR, Fiedler U, Span S, Lingnau A, Pham T V., Hoffmann S, et al. (2010) **Workflow comparison for label-free, quantitative secretome proteomics for cancer biomarker discovery: Method evaluation, differential analysis, and verification in serum** *J Proteome Res* **9**:1913–22

30. Cox J, Mann M (2008) **MaxQuant enables high peptide identification rates, individualized p.p.b.-range mass accuracies and proteome-wide protein quantification** *Nat Biotechnol* **26**:1367–72
31. Raudvere U, Kolberg L, Kuzmin I, Arak T, Adler P, Peterson H, et al. (2019) **g:Profiler: a web server for functional enrichment analysis and conversions of gene lists (2019 update)** *Nucleic Acids Res* **47**:W191–8
32. Subramanian A, Tamayo P, Mootha VK, Mukherjee S, Ebert BL, Gillette MA, et al. (2005) **Gene set enrichment analysis: A knowledge-based approach for interpreting genome-wide expression profiles** *Proc Natl Acad Sci U S A* **102**:15545–50
33. Sergushichev AA (2016) **An algorithm for fast preranked gene set enrichment analysis using cumulative statistic calculation** *bioRxiv* **60012**
34. Liberzon A, Subramanian A, Pinchback R, Thorvaldsdóttir H, Tamayo P, Mesirov JP (2011) **Molecular signatures database (MSigDB) 3.0** *Bioinformatics* **27**:1739–40
35. Hänzelmann S, Castelo R, Guinney J (2013) **GSVA: Gene set variation analysis for microarray and RNA-Seq data** *BMC Bioinformatics* **14**
36. Gillespie M, Jassal B, Stephan R, Milacic M, Rothfels K, Senff-Ribeiro A, et al. (2022) **The reactome pathway knowledgebase 2022** *Nucleic Acids Res* **50**:D687–92
37. Consortium TGO, Aleksander SA, Balhoff J, Carbon S, Cherry JM, Drabkin HJ, et al. (2023) **The Gene Ontology knowledgebase in 2023** *Genetics* **224**
38. Ashburner M, Ball CA, Blake JA, Botstein D, Butler H, Cherry JM, et al. (2000) **Gene Ontology: tool for the unification of biology** *Nat Genet* **25**:25–9
39. Beekhof R, Alphen C van, Henneman AA, Knol JC, Pham T V, Rolfs F, et al. (2019) **INKA, an integrative data analysis pipeline for phosphoproteomic inference of active kinases** *Mol Syst Biol* **15**
40. Doskey CM, Erve TJ van 't, Wagner BA, Buettner GR. (2015) **Moles of a Substance per Cell Is a Highly Informative Dosing Metric in Cell Culture** *PLoS One* **10**
41. Chen Q, Espey MG, Krishna MC, Mitchell JB, Corpe CP, Buettner GR, et al. (2005) **Pharmacologic ascorbic acid concentrations selectively kill cancer cells: Action as a pro-drug to deliver hydrogen peroxide to tissues** *Proc Natl Acad Sci U S A* **102**:13604–13609
42. Kirtonia A, Sethi G, Garg M (2020) **The multifaceted role of reactive oxygen species in tumorigenesis** *Cell Mol Life Sci* **77**:4459–83
43. Brown RAM, Richardson KL, Kabir TD, Trinder D, Ganss R, Leedman PJ (2020) **Altered Iron Metabolism and Impact in Cancer Biology, Metastasis, and Immunology** *Front Oncol* **10**
44. Lane DJR, Richardson DR. (2014) **The active role of vitamin C in mammalian iron metabolism: Much more than just enhanced iron absorption!** *Free Radic Biol Med* **75**:69–83
45. Soula M, Weber RA, Zilka O, Alwaseem H, La K, Yen F, et al. (2020) **Metabolic determinants of cancer cell sensitivity to canonical ferroptosis inducers** *Nat Chem Biol* **16**:1351–60

46. Frejno M, Chiozzi RZ, Wilhelm M, Koch H, Zheng R, Klaeger S, et al. (2017) **Pharmacoproteomic characterisation of human colon and rectal cancer** *Mol Syst Biol* **13**
47. Linnekamp JF, Van Hooff SR, Prasetyanti PR, Kandimalla R, Buikhuisen JY, Fessler E, et al. (2018) **Consensus molecular subtypes of colorectal cancer are recapitulated in in vitro and in vivo models** *Cell Death Differ* **25**:616–33
48. Tanner JJ, Fendt SM, Becker DF (2018) **The Proline Cycle As a Potential Cancer Therapy Target** *Biochemistry* **57**:3433–44
49. Huang H, Zhang S, Li Y, Liu Z, Mi L, Cai Y, et al. (2021) **Suppression of mitochondrial ROS by prohibitin drives glioblastoma progression and therapeutic resistance** *Nat Commun* **12**
50. Miles AL, Burr SP, Grice GL, Nathan JA (2017) **The vacuolar-ATPase complex and assembly factors, TMEM199 and CCDC115, control HIF1 α prolyl hydroxylation by regulating cellular iron levels** *Elife* **6**
51. Dai X, Cheng H, Bai Z, Li J (2017) **Breast Cancer Cell Line Classification and Its Relevance with Breast Tumor Subtyping** *J Cancer* **8**
52. Huang L, Xu AM, Liu W (2015) **Transglutaminase 2 in cancer** *Am J Cancer Res* **5**
53. Costantini S, Capone F, Polo A, Bagnara P, Budillon A (2021) **Valosin-Containing Protein (VCP)/p97: A Prognostic Biomarker and Therapeutic Target in Cancer** *Int J Mol Sci* **22**
54. Michael A, Ainsley A, Joseph A, Jahan N, Michael A, Ainsley A, et al. (2020) **First and Second Line Chemotherapeutic Regimens for Non-Small Cell Lung Carcinomas - The Efficacy of Platinum, Non-Platinum and Combination Therapy: A Literature Review** *Cureus* **12**
55. Cortés ÁA, Urquizu LC, Cubero JH (2015) **Adjuvant chemotherapy in non-small cell lung cancer: state-of-the-art** *Transl lung cancer Res* **4**:191–7
56. Böttger F, Schaaij-Visser TB, de Reus I, Piersma SR, Pham T V., Nagel R, et al. (2019) **Proteome analysis of non-small cell lung cancer cell line secretomes and patient sputum reveals biofluid biomarker candidates for cisplatin response prediction** *J Proteomics* **196**:106–19
57. Sorenson GD, Pettengill S, Brinck-Johnsen T, Cate CC, Maurer LH (1981) **Hormone Production by Cultures of Small-Cell Carcinoma of the Lung** *Cancer*
58. Kranz RG, Richard-Fogal C, Taylor J-S, Frawley ER (2009) **Cytochrome c Biogenesis: Mechanisms for Covalent Modifications and Trafficking of Heme and for Heme-Iron Redox Control** *Microbiol Mol Biol Rev* **73**:510–28
59. Pin G, Huanting L, Chengzhan Z, Xinjuan K, Yugong F, Wei L, et al. (2020) **Down-Regulation of PDCD4 Promotes Proliferation, Angiogenesis and Tumorigenesis in Glioma Cells** *Front Cell Dev Biol* **8**
60. Ali A, Vineethakumari C, Lacasa C, Lüders J (2023) **Microtubule nucleation and γ TuRC centrosome localization in interphase cells require ch-TOG** *Nat Commun* **14**:1–17
61. Miller MP, Asbury CL, Biggins S (2016) **A TOG Protein Confers Tension Sensitivity to Kinetochore-Microtubule Attachments** *Cell* **165**:1428–39

62. Kim Y, Kim H, Bae S, Choi J, Lim SY, Lee N, et al. (2013) **Vitamin C Is an Essential Factor on the Anti-viral Immune Responses through the Production of Interferon- α/β at the Initial Stage of Influenza A Virus (H3N2) Infection** *Immune Netw* **13**
63. Ho WJ, Jaffee EM, Zheng L (2020) **The tumour microenvironment in pancreatic cancer — clinical challenges and opportunities** *Nat Rev Clin Oncol* **17**:527–40
64. Du W, Menjivar RE, Donahue KL, Kadiyala P, Velez-Delgado A, Brown KL, et al. (2023) **WNT signaling in the tumor microenvironment promotes immunosuppression in murine pancreatic cancer** *J Exp Med* **220**
65. Kim JH, Hwang S, Lee JH, Im SS, Son J (2022) **Vitamin C Suppresses Pancreatic Carcinogenesis through the Inhibition of Both Glucose Metabolism and Wnt Signaling** *Int J Mol Sci* **23**
66. Garrido C, Galluzzi L, Brunet M, Puig PE, Didelot C, Kroemer G (2006) **Mechanisms of cytochrome c release from mitochondria** *Cell Death Differ* **13**:1423–33
67. Lee EE, Ma J, Sacharidou A, Mi W, Salato VK, Nguyen N, et al. (2014) **A Protein Kinase C Phosphorylation Motif in GLUT1 Affects Glucose Transport and is Mutated in GLUT1 Deficiency Syndrome** *Mol Cell* **58**:845–53
68. Lee S, Chang J, Blackstone C (2016) **FAM21 directs SNX27-retromer cargoes to the plasma membrane by preventing transport to the Golgi apparatus** *Nat Commun* **7**
69. Kearney AL, Cooke KC, Norris DM, Zadoorian A, Krycer JR, Fazakerley DJ, et al. (2019) **Serine 474 phosphorylation is essential for maximal Akt2 kinase activity in adipocytes** *J Biol Chem* **294**:16729–39
70. Mustafi S, Camarena V, Qureshi R, Sant DW, Wilkes Z, Bilbao D, et al. (2021) **Vitamin C sensitizes triple negative breast cancer to PI3K inhibition therapy** *Theranostics* **11**
71. Erickson JR, Julie He B, Grumbach IM, Anderson ME (2011) **CaMKII in the cardiovascular system: Sensing redox states** *Physiol Rev* **91**:889–915
72. Chen Q, Espey MG, Sun AY, Pooput C, Kirk KL, Krishna MC, et al. (2008) **Pharmacologic doses of ascorbate act as a prooxidant and decrease growth of aggressive tumor xenografts in mice** *Proc Natl Acad Sci* **105**:11105–9
73. Xu J, Jia Z, Knutson MD, Leeuwenburgh C (2012) **Impaired Iron Status in Aging Research .** *Int J Mol Sci* **13**:2368–86
74. Chen X, Kang R, Kroemer G, Tang D (2021) **Organelle-specific regulation of ferroptosis** *Cell Death Differ* **28**:2843–56
75. Wang X, Xu S, Zhang L, Cheng X, Yu H, Bao J, et al. (2021) **Vitamin C induces ferroptosis in anaplastic thyroid cancer cells by ferritinophagy activation** *Biochem Biophys Res Commun* **551**:46–53
76. Liu Y, Huang P, Li Z, Xu C, Wang H, Jia B, et al. (2022) **Vitamin C Sensitizes Pancreatic Cancer Cells to Erastin-Induced Ferroptosis by Activating the AMPK/Nrf2/HMOX1 Pathway** *Oxid Med Cell Longev* **2022**

77. El Hassouni B, Granchi C, Vallés-Martí A, Supadmanaba IGP, Bononi G, Tuccinardi T, et al. (2020) **The dichotomous role of the glycolytic metabolism pathway in cancer metastasis: Interplay with the complex tumor microenvironment and novel therapeutic strategies** *Semin Cancer Biol* **60**:238–48
78. Goel C, Monga SP, Nejak-Bowen K (2022) **Role and Regulation of Wnt/ β -Catenin in Hepatic Perivenous Zonation and Physiological Homeostasis** *Am J Pathol* **192**:4–17
79. Qin S, Wang G, Chen L, Geng H, Zheng Y, Xia C, et al. (2023) **Pharmacological vitamin C inhibits mTOR signaling and tumor growth by degrading Rictor and inducing HMOX1 expression** *PLOS Genet* **19**
80. Bakalova R, Zhelev Z, Miller T, Aoki I, Higashi T (2020) **Vitamin C versus Cancer: Ascorbic Acid Radical and Impairment of Mitochondrial Respiration?** *Oxid Med Cell Longev* **2020**
81. Pizzo V, Lendeckel U, Karimi F, Al Abdulla R, Wolke C (2023) **The Role of the Ectopeptidase APN/CD13 in Cancer** *Biomed* **11**
82. Welsh JL, Wagner BA, van't Erve TJ, Zehr PS, Berg DJ, Halfdanarson TR, et al. (2013) **Pharmacological ascorbate with gemcitabine for the control of metastatic and node-positive pancreatic cancer (PACMAN): results from a phase I clinical trial** *Cancer Chemother Pharmacol* **71**:765–75
83. Polireddy K, Dong R, Reed G, Yu J, Chen P, Williamson S, et al. (2017) **High Dose Parenteral Ascorbate Inhibited Pancreatic Cancer Growth and Metastasis: Mechanisms and a Phase I/IIa study** *Sci Rep* **7**
84. Perez-Riverol Y, Bai J, Bandla C, García-Seisdedos D, Hewapathirana S, Kamatchinathan S, et al. (2022) **The PRIDE database resources in 2022: a hub for mass spectrometry-based proteomics evidences** *Nucleic Acids Res* **50**

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Reviewer #1 (Public Review):

Summary:

The study by Valles-Martí et al. was aimed at elucidating mechanisms of high-dose vitamin C (Ascorbate) sensitivity using proteomics of a large panel of cancer cell lines. The study is primarily based on correlating protein expression to vitamin C sensitivity based on IC50 from cell viability studies. As expected, cancer type-specific proteome patterns emerge and the authors conclude that some pan-cancer pathways, such as proliferation correlate with high

sensitivity to VitC. In a subset of PDAC cells proteomics and phospho proteomics were also carried out following vitamin C treatment, albeit those studies did not identify significant changes in response to treatment.

Strengths:

The premise for the work is of interest as high dose vitamin C is in clinical trials and thus studies investigating mechanisms of sensitivity and potential resistance mechanisms to this therapy are of interest to the field. The authors have collected large proteomic datasets on some of the most common cancer cells used and these data may be a useful resource for others when made publicly available. Although this is not necessarily novel, since proteomics data sets for some of the included cell lines are already available.

Weaknesses:

The title suggests that the proteomics data presented "underscores high-dose vitamin C as a potent anti-cancer agent" However, while the proteomic data are extensive, it is my assessment that without further validation there are no clear pathways identified by the presented proteomics data that conclusively determine vitamin C sensitivity.

A major question arising from this work is how specific the proteomics data reflect sensitivity to vitamin C over general sensitivity to other cytotoxic agents. It would be of interest to compare the correlation of proteomic data and ascorbate sensitivity to the sensitivity of cell lines to other cytotoxic agents. (e.g. comparison to NCI-60 growth inhibition data). In other words, do the proteomic data that correlate with ascorbate sensitivity simply reflect susceptibility to other cytotoxic agents? The comments that vitamin C toxicity is not dependent on underlying histological or genetic subtypes of cancers ("one size fits all") suggest this.

The genetic backgrounds of tumor cells have not been taken into consideration in the analysis and how this may influence VitC susceptibility. An example that comes to mind is KEAP1/Nrf2 aberrations in lung cancer.

The study would be significantly strengthened if some of the proteins identified were further validated in eliciting low or high sensitivity to Vitamin C. Of particular interest are proteins that have functions related to known mechanisms of action of Vitamin C toxicity, such as iron homeostasis. Some of the metabolic-related protein changes are also of interest. For example, HCCS expression is mentioned several times as being associated with lower sensitivity to ascorbate. Providing experimental evidence that this protein is of significance to Vitamin C sensitivity and if this is due to its effects on iron and subsequent generation of ROS in response to VitC would be of significance.

Similarly, an interesting aspect of the findings is the authors' conclusion that proliferation is associated with Vitamin C sensitivity. The authors propose in their discussion that Vitamin C may be an attractive alternative to treat heavily pretreated and chemoresistant cancers. Thus it would be important to know which of the highly proliferative cell lines tested have a chemoresistance phenotype and are also more susceptible to Vitamin C toxicity. Perhaps partitioning the cells further into chemoresistant and sensitive cell lines to standard chemotherapy and then assessing which protein signatures are associated with Vitamin C sensitivity will allow for better elucidation of sensitivity mechanisms that are more relevant to using Vitamin C as an alternate therapy for chemoresistant tumors.

Following on from this, there is an interesting mechanistic question as to why more proliferative cells are more sensitive to vitamin C, and whether this is related to changes in metabolism and underlying changes in their steady-state levels of ROS. Further investigating this mechanistically based on the identified proteomic signatures could make the findings more significant.

Vitamin C can also generate H₂O₂ extracellularly in the presence of iron. Thus, Vitamin C toxicity could be affected by different abilities of the tumor cells to scavenge extracellular H₂O₂, such as different expression levels of extracellular antioxidant enzymes. Judging from the methods section, it does not appear that proteomic data include secreted proteins. Can the authors comment on how this may be a potential caveat?

In light of this, the strong effects of exogenous catalase addition on cell viability suggest that H₂O₂ may be produced by ascorbate in the media.

Similarly, can the authors comment on the cell culture conditions used to compare IC₅₀s between cell lines, specifically if different media and FBS batches were used, as these have the potential to vary in metal/iron concentrations that might influence the pro-oxidant generation by high dose ascorbate in media. Specifically, have the authors looked into the iron content and how these different conditions may be contributing to intracellular H₂O₂ and extracellular H₂O₂ (AmplexRed) production in response to Vitamin C.

Other comments relate to methods:

How was ascorbate prepared? There is no mention of degassing of H₂O and ensuring that H₂O does not have metal impurities, which can lead to auto-oxidation.

The OxiSelect probe is based on DCFDA, which is an oxidant-sensitive probe that has been described to be fraught with artifacts. Thus it is advised to mention the caveats associated with the use of this probe (as outlined in PMID: PMC3911769) and consider backing up these experiments with additional Oxidant probes.

<https://doi.org/10.7554/eLife.94988.1.sa2>

Reviewer #2 (Public Review):

Summary:

The authors generated proteome profiles of 51 cancer cell lines treated with pharmacologic ascorbate. The idea was to identify players responsible for the sensitivity or relative resistance to ascorbate to delineate mechanisms of action of this potentially transformative new treatment.

Strengths:

The proteomic profiles themselves. The identification of MAPK and mTOR as overrepresented proteomic elements and close correlations between proliferation, cell cycle mediators, and sensitivity to ascorbate indicate that rapidly proliferating cancer may be more sensitive to ascorbate. Also, the finding that sensitivity to ascorbate is correlated to different pathways in different types of cancer is interesting. For instance, in some pancreatic and lung cancers sensitivity seemed to be related to iron handling while in breast DNA damage/repair seemed to be most involved.

Weaknesses:

The study is quite descriptive. Although the proteomes indicate what pathways are more or less represented after ascorbate challenge there is little mechanistic information about their relevance to the sensitivity to ascorbate. Since activity is not assessed, proteins may be present in higher or lower abundance but not necessarily at the peak of their activity. Also, many statements are made as "known facts" but no references are provided.

<https://doi.org/10.7554/eLife.94988.1.sa1>

Reviewer #3 (Public Review):**Summary:**

In the manuscript titled "Large pan-cancer cell screen coupled to (phospho-)proteomics underscores high-dose vitamin C (VitC) as a potent anti-cancer agent," the authors use a combination of proteomics and cell viability assays to understand the effect of Vitamin C on different solid tumor models in 51 different cancer cell lines. They found that many cancer cell lines are sensitive to high-dose Vitamin C, with IC50 values in the micromolar to millimolar range. Given that Vitamin C, when administered intravenously, can reach 20mM, this suggests that Vitamin C could provide some benefits to patients. The authors also generate and analyze bulk proteomic data for all 51 cell lines. They perform statistical analysis of these data to identify proteins that are up or downregulated in sensitive vs resistant cell lines in the same tumor and commonly across tumors. They then focus on PDAC cell lines and measure bulk and phosphoproteomics of PDAC cell lines 2, 4 and 24 hours after Vitamin C treatment.

Strengths: The strengths of the study are the rather large datasets accumulated on bulk proteomics of 51 different cancer cell lines. The IC50 values of these cell lines in response to Vitamin C is also useful.

Weaknesses:

Though identifying targets to sensitize cancer cells to Vitamin C treatment is interesting, I felt the manuscript delved too much into listing off genes they found, with speculation on why the particular protein would be enriched in sensitive or resistant cell lines without testing any key claims experimentally.

Major Issues

(1) The overall premise of the study is that proteins that are enriched in Vitamin C-sensitive cell lines point to mechanisms of sensitivity and those enriched in Vitamin C resistant lines underlie mechanisms of resistance. Yet this is never directly tested. To show that the authors would need to knockdown/knockout a gene enriched in resistant lines and show this sensitizes cells to Vitamin C treatment or overexpress a protein associated with resistance and show that this leads to resistance in an otherwise sensitive cell line.

(2) One of the key strengths of this study is the large datasets generated, namely the proteomics data for 51 different cell lines. Yet the data is not included as a supplement or uploaded to a public repository.

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